IMOTC 2026

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§1 Invariants

See [FGI96, Chapter 12], [Eng98, Chapter 1].

§1.1 Impossibility

Exercise 1 (ELMO 1999 P2,). Mr. Fat moves around on the lattice points according to the following rules: From point (x, y) he may move to any of the points (y, x), $(3x, -2y)$, $(-2x, 3y)$, $(x+1, y+4)$ and $(x-1, y-4)$. Show that if he starts at $(0, 1)$ he can never get to $(0, 0)$.

> **Walkthrough** — Show that under the given moves, the modulo 5 congruence class of the sum of the coordinates of the point where Mr. Fat is located remains invariant upto a factor of 3.

Solution 1. Note that if he moves from (x, y) to (a, b), then one of the congruences

$$a + b \equiv x + y \pmod 5, a + b \equiv 3(x + y) \pmod 5$$

hold. Since the sum of the coordinates of $(0, 1)$ is nonzero modulo 5, and the sum of the coordinates of $(0, 0)$ is zero, it follows that if he starts at $(0, 1)$ he can never get to $(0, 0)$. ■

Exercise 2 (RMO 2014b P6,). [Eng98, Problem 28, p. 29, p. 35] Let n be an odd positive integer and suppose that each square of an $n \times n$ grid is arbitrarily filled with either by 1 or by -1. Let r_j and c_k denote the product of all numbers in j-th row and k-th column respectively, $1 \leq j, k \leq n$. Prove that

$$\sum_{j=1}^{n} r_j + \sum_{k=1}^{n} c_k \neq 0.$$

Walkthrough —

(a) What would happen to the sum if the sign of one of the entries is changed? Does the sum change modulo 4?

(b) What would happen when all the entries are changed to 1 by changing the signs of the negative entries?

Solution 2. Denote the sum $\sum_{j=1}^{n} r_j + \sum_{k=1}^{n} c_k$ by S. If all entries of the grid are equal to 1, then S is equal to $2n$. Note that if the sign on an entry of the grid is changed, then S does not change modulo 4. So the integer S is congruent to $2n$ modulo 4. Since n is odd, the integer S is nonzero. For an alternate solution, see [Eng98, Problem 29, Chapter 2, p. 28]. ■

Exercise 3 (Cono Sur Olympiad 2013 P1,). Four distinct points are marked in a line. For each point, the sum of the distances from said point to the other three is calculated; getting in total 4 numbers. Decide whether these 4 numbers can be, in some order:

(i) $29, 29, 35, 37$,

(ii) $28, 29, 35, 37$,

(iii) $28, 34, 34, 37$.

Walkthrough —

(a) For each of the four points, express the sum of the distances to the other three points in terms of the distances between consecutive points among the four points.

(b) Show that at least two of the four numbers must be equal.

(c) How does it follow that the first set of numbers is the only one that can be obtained?

Solution 3. Denote the points by A, B, C, D from left to right. Let x, y, z be the distances between consecutive points, that is, $x = AB, y = BC, z = CD$. For P in $\{A, B, C, D\}$, let S_P be the sum of the distances from P to the other three points. Then, we have

$$\begin{aligned} S_A &= AB + AC + AD \\ &= x + (x + y) + (x + y + z) \\ &= 3x + 2y + z, \\ S_B &= AB + BC + BD \\ &= x + y + (y + z) \\ &= x + 2y + z, \end{aligned}$$

$$\begin{aligned} S_C &= AC + BC + CD \\ &= (x + y) + y + z \\ &= x + 2y + z, \\ S_D &= AD + BD + CD \\ &= (x + y + z) + (y + z) + z \\ &= x + 2y + 3z. \end{aligned}$$

Note that $S_B = S_C$. Thus, at least two of the numbers S_A, S_B, S_C, S_D are equal. This is only the case for the first set of numbers, $29, 29, 35, 37$. Also observe that for $x = 3, y = 11, z = 4$, we have $S_A = 35, S_B = S_C = 29, S_D = 37$. This shows that only the first set of numbers can occur as the sums of the distances from the four points to the other three points in some order. ■

Exercise 4 (Japan Mathematical Olympiad 2018 P1,). Positive integers between 1 to 100 inclusive are written on a blackboard, each exactly once. One operation involves choosing two numbers a and b on the blackboard and erasing them, then writing the greatest common divisor of $a^2 + b^2 + 2$ and $a^2b^2 + 3$. After a number of operations, there is only one positive integer left on the blackboard. Prove this number cannot be a perfect square.

Walkthrough —

(a) Show that the parity of the number of multiples of 3 on the blackboard is an invariant.

(b) Show that the last number on the blackboard is a multiple of 3, but is not a multiple of 9.

Solution 4.

Claim — The parity of the number of multiples of 3 on the blackboard is an invariant.

Proof of the claim. Let a and b be the two numbers chosen in an operation. If a and b are both multiples of 3, then $a^2 + b^2 + 2$ is not divisible by 3, and hence, $\gcd(a^2 + b^2 + 2, a^2b^2 + 3)$ is not divisible by 3. If exactly one of a and b is a multiple of 3, then $a^2 + b^2 + 2$ and $a^2b^2 + 3$ are both divisible by 3, and hence, $\gcd(a^2 + b^2 + 2, a^2b^2 + 3)$ is divisible by 3. If neither a nor b is a multiple of 3, then $a^2 + b^2 + 2$ is not divisible by 3, and hence, $\gcd(a^2 + b^2 + 2, a^2b^2 + 3)$ is not divisible by 3. Therefore, the parity of the number of multiples of 3 on the blackboard remains unchanged after each operation. □

Initially, there are 33 multiples of 3 on the blackboard, so the parity of the number of multiples of 3 is odd. After performing 99 operations, there is only one number left on the blackboard, and it must be a multiple of 3. Let x, y be

the last two numbers chosen in the final operation. Then, the last number on the blackboard is $\gcd(x^2+y^2+2, x^2y^2+3)$. This shows that 3 divides x^2y^2+3, which implies that 3 divides xy. Therefore, 9 does not divide x^2y^2+3, and hence, 9 does not divide $\gcd(x^2+y^2+2, x^2y^2+3)$. This shows that the last number on the blackboard is divisible by 3, but is not divisible by 9, and hence is not a perfect square. ■

Exercise 5 (Tournament of Towns 2021 Spring Junior A Level P2, , proposed by Maxim Didin)**.** In a room there are several children and a pile of 1000 sweets. The children come to the pile one after another in some order. Upon reaching the pile each of them divides the current number of sweets in the pile by the number of children in the room, rounds the result if it is not integer, takes the resulting number of sweets from the pile and leaves the room. All the boys round upwards and all the girls round downwards. The process continues until everyone leaves the room. Prove that the total number of sweets received by the boys does not depend on the order in which the children reach the pile.

The following problem is similar to Exercise 9.

Exercise 6 (Singapore Junior Mathematical Olympiad 2019 Round 2 P2,)**.** There are 315 marbles divided into three piles of 81, 115 and 119. In each move Ah Meng can either merge several piles into a single pile or divide a pile with an even number of marbles into 2 equal piles. Can Ah Meng divide the marbles into 315 piles, each with a single marble?

> **Walkthrough** —
>
> **(a)**

Solution 5. to write. ■

Exercise 7 (Belarus National Olympiad 2023 Grade 8 Day 1 P1,)**.** A move on an unordered triple of numbers (a, b, c) changes the triple to either $(a, b, 2a+2b-c)$, $(a, 2a+2c-b, c)$ or $(2b+2c-a, b, c)$. Can you perform a finite sequence of moves on the triple $(3, 5, 14)$ to get the triple $(3, 13, 6)$?

> **Walkthrough** — Show that if a sequence of moves is performed on a triple (a, b, c), then the resulting **unordered** triple is congruent to one of the triples
>
> $$(a, b, -(a+b+c)), (a, -(a+b+c), c), (-(a+b+c), b, c), (a, b, c)$$
>
> modulo 3, that is, the residues of the entries of the resulting triple modulo 3 coincide with the residues of the entries of one of the above triples in some order.

Solution 6. In the following, any triple is considered as an **unordered** triple.

> **Claim** — If a sequence of moves is performed on an **unordered** triple (a, b, c), then the resulting **unordered** triple is congruent to one of the **unordered** triples
>
> $$(a, b, -(a+b+c)), (a, -(a+b+c), c), (-(a+b+c), b, c), (a, b, c)$$
>
> modulo 3.

Proof of the Claim. It suffices to prove that an admissible move on any of the triples

$$(a, b, -(a+b+c)), (a, -(a+b+c), c), (-(a+b+c), b, c), (a, b, c)$$

results in a triple which is congruent to one of the above triples modulo 3. Note that applying an admissible move to the triple $(a, b, -(a+b+c))$ results in a triple which is congruent to one of the triples

$$(c, a, b), (c, a, -(a+b+c)), (c, b, -(a+b+c))$$

modulo 3. Similarly, applying an admissible move to the triple $(a, -(a+b+c), c)$ results in a triple which is congruent to one of the triples

$$(b, a, c), (b, a, -(a+b+c)), (b, c, -(a+b+c))$$

modulo 3. Applying an admissible move to the triple $(-(a+b+c), b, c)$ results in a triple which is congruent to one of the triples

$$(a, b, c), (a, b, -(a+b+c)), (a, c, -(a+b+c))$$

modulo 3. Finally, applying an admissible move to the triple (a, b, c) results in a triple which is congruent to one of the triples

$$(a, b, -(a+b+c)), (a, c, -(a+b+c)), (b, c, -(a+b+c))$$

modulo 3. In each case, the resulting triple is congruent to one of the triples

$$(a, b, -(a+b+c)), (a, -(a+b+c), c), (-(a+b+c), b, c), (a, b, c)$$

modulo 3. Thus, the claim is proved. □

Note that the triple $(3, 5, 14)$ is congruent to the triple $(0, 2, 2)$ modulo 3. By the claim, if a sequence of moves is performed on the triple $(3, 5, 14)$, then the resulting triple is congruent to one of the triples

$$(2, 2, 2), (0, 2, 2)$$

modulo 3. The triple $(3, 13, 6)$ is congruent to the triple $(0, 1, 0)$ modulo 3. Since $(0, 1, 0)$ is not congruent to either $(2, 2, 2)$ or $(0, 2, 2)$ modulo 3, it is impossible to perform a finite sequence of moves on the triple $(3, 5, 14)$ to get the triple $(3, 13, 6)$. ■

Remark. Note that $a^2+b^2+c^2-ab-bc-ca$ is an invariant.

Exercise 8 (Belarus National Olympiad 2023 Grade 9 Day 1 P2,). A move on an unordered triple of numbers (a,b,c) changes the triple to either $(a,b,2a+2b-c)$, $(a,2a+2c-b,c)$ or $(2b+2c-a,b,c)$. Can you perform a finite sequence of moves on the triple $(3,5,14)$ to get the triple $(9,8,11)$?

Walkthrough — Show that the mod 4 congruence class of the sum of the entries of such a triple remains invariant under the allowed moves if the sum of the entries of the initial triple is congruent to 2 modulo 4.

Solution 7. Note that for any unordered triple (a,b,c), the sum of its entries is congruent to the sum of the entries of any triple obtained from (a,b,c) by an admissible move modulo 4 if the sum of the entries of the initial triple is congruent to 2 modulo 4. Indeed, the sum of the entries of any of the triples $(a,b,2a+2b-c)$, $(a,2a+2c-b,c)$ and $(2b+2c-a,b,c)$ is congruent to $-(a+b+c)$ modulo 4. Hence, if the sum of the entries of the initial triple is congruent to 2 modulo 4, then the sum of the entries of any triple obtained from it by a finite sequence of moves is also congruent to 2 modulo 4. Since the sum of the entries of the triple $(3,5,14)$ is 22, which is congruent to 2 modulo 4, and the sum of the entries of the triple $(9,8,11)$ is 28, which is congruent to 0 modulo 4, it is impossible to perform a finite sequence of moves on the triple $(3,5,14)$ to get the triple $(9,8,11)$. ■

Remark. Note that $a^2+b^2+c^2-ab-bc-ca$ is an invariant.

Exercise 9 (Kosovo National Olympiad 2023 Grade 11 P1,). In three different piles, there are 51, 49 and 5 stones, respectively. You can combine two piles in a larger pile or you can separate a pile that contains an even amount of stones into two equal piles. Is it possible that after some turns, we can separate the stones into 105 piles with 1 stone on each pile?

Walkthrough —

(a) Show that the greatest common divisor of the numbers of stones of the piles is an integer larger than 1.

(b) Show that it is impossible to separate the stones into 105 piles with 1 stone on each pile.

Solution 8. Note that if the greatest common divisor of the numbers of stones of the piles before some turn is an integer d, then d divides the number of stones

in each pile after that turn. Since $5, 49, 51$ are all odd, it follows that during the first turn two piles are to be combined. Hence, the numbers of stones in the piles after the first turn are equal to $5, 100$, or to $49, 56$, or to $51, 54$. This shows that the greatest common divisor of the numbers of stones of the piles is an integer larger than 1. Consequently, it is impossible to separate the stones into 105 piles with 1 stone on each pile after some turns. ■

Exercise 10 (Canadian Mathematical Olympiad 2024 P2,). Jane writes down 2024 natural numbers around the perimeter of a circle. She wants the 2024 products of adjacent pairs of numbers to be exactly the set $\{1!, 2!, \ldots, 2024!\}$. Can she accomplish this?

> **Walkthrough** —
>
> **(a)** Show that to have such a configuration, the product $1! \cdot 2! \cdot \ldots \cdot 2024!$ must be a perfect square.
>
> **(b)** Show that $1! \cdot 2! \cdot \ldots \cdot 2024!$ is not a perfect square.

Solution 9. We claim that Jane cannot accomplish this. Assume for the sake of contradiction that she can. Let the numbers be $a_1, a_2, \ldots, a_{2024}$ in clockwise order, and let $a_{2025} = a_1$. Then for some permutation $k_1, k_2, \ldots, k_{2024}$ of $1, 2, \ldots, 2024$, we have $a_i a_{i+1} = k_i!$ for any $1 \leq i \leq 2024$. It follows that

$$(a_1 a_2 a_3 \cdots a_{2024})^2 = 1! \cdot 2! \cdot \ldots \cdot 2024!.$$

Note that 1009 divides each of $1009!, 1010!, \ldots, 2024!$ but does not divide any of $1!, 2!, \ldots, 1008!$. Moreover, 1009^2 does not divide any of $1009!, 1010!, \ldots, 2017!$. Also note that 1009^2 divides $2018!, 2019!, \ldots, 2024!$, and 1009^3 does not divide any of $2018!, 2019!, \ldots, 2024!$. This implies that the largest nonnegative integer r such that 1009^r divides $1! \cdot 2! \cdot \ldots \cdot 2024!$ is equal to

$$(2024 - 1008) + (2024 - 2017),$$

which is odd. This implies that $1! \cdot 2! \cdot \ldots \cdot 2024!$ is not a perfect square, contradicting the earlier equation. Hence, Jane cannot accomplish this. ■

> **Remark.** Note that no prime larger than 1009 helps to show that $1! \cdot 2! \cdot \ldots \cdot 2024!$ is not a perfect square.

§1.2 Determining certain quantities

Example 11. Suppose the numbers $1, \frac{1}{2}, \frac{1}{3}, \ldots, \frac{1}{100}$ are written on a blackboard. In each step, you may erase two numbers a and b and replace them with the number $a + b + ab$. After 99 steps, only one number remains. What are the possible values of this number?

Walkthrough —

(a)

Solution 10. For any two rationals a and b, define

$$a \oplus b = a + b + ab = (a+1)(b+1) - 1,$$

and call $a \oplus b$ the *nice sum* of a and b. Note that the operation $\oplus$ is associative and commutative. Hence, we may consider the nice sum of all the numbers on the blackboard, which is well-defined. The nice sum of a single number a is defined as $a \oplus 0 = a$. Note that the nice sum of the numbers present on the blackboard at a given time does not change after a step, since the two numbers a and b are replaced by their nice sum $a \oplus b$. Initially, the nice sum of the numbers on the blackboard is

$$(1+1)\left(\frac{1}{2}+1\right)\left(\frac{1}{3}+1\right)\cdots\left(\frac{1}{100}+1\right) - 1 = 101 - 1 = 100.$$

Therefore, the nice sum of the numbers on the blackboard is always 100. In particular, the nice sum of the final number is 100, so the final number is 100 regardless of the choices made in the steps. ■

Exercise 12 (Argentina National Olympiad 2018 Level 2 P1,). A list of 2018 numbers is created using the following procedure: the first number is 47, the second number is 74, and from there, each number is equal to the number formed by the last two digits of the sum of the two previous numbers:

$$47, 74, 21, 95, 16, 11, \ldots$$

Bruno squares each of the 2018 numbers and sums them. Determine the remainder when this sum is divided by 8.

Walkthrough —

(a) Consider the residues of the numbers in the list modulo 4.

(b) Determine the residues of the squares of the numbers in the list modulo 8.

(c) Determine the remainder when the sum of the squares of the numbers in the list is divided by 8.

Solution 11. Note that the first two integers on the list are congruent to 3 and 2 modulo 4, respectively. Therefore, the next six integers on the list are congruent to 1, 3, 0, 3, 3, and 2 modulo 4, respectively. Applying induction, it follows that the residues of the numbers in the list modulo 4 are periodic with period 6:

$$3, 2, 1, 3, 0, 3, 3, 2, 1, 3, 0, 3, \ldots.$$

Hence, the residues of the squares of the numbers in the list modulo 8 are also periodic with period 6:

$$1, 4, 1, 1, 0, 1, 1, 4, 1, 1, 0, 1, \dots.$$

Since $2018 = 6 \cdot 336 + 2$, the sum of the squares of the numbers in the list is congruent to

$$336 \cdot (1 + 4 + 1 + 1 + 0 + 1) + 1 + 4$$

modulo 8. Therefore, the remainder when this sum is divided by 8 is 5. ■

Exercise 13 (Balkan Mathematical Olympiad 2018 P2, , proposed by Jeremy King, UK)**.** Let q be a positive rational number. Two ants are initially at the same point X in the plane. In the n-th minute ($n = 1, 2, \dots$) each of them chooses whether to walk due north, east, south or west and then walks the distance of q^n metres. After a whole number of minutes, they are at the same point in the plane (not necessarily X), but have not taken exactly the same route within that time. Determine all possible values of q.

Walkthrough —

(a) Let q be a positive rational number such that the two ants can be at the same point in the plane after a whole number of minutes, without taking the same route.

(b) Let r be a positive integer such that the two ants are at the same point in the plane after r minutes, without taking the same route.

(c) Prove that

$$\varepsilon_1 q + \varepsilon_2 q^2 + \cdots + \varepsilon_r q^r = \varepsilon_1' q + \varepsilon_2' q^2 + \cdots + \varepsilon_r' q^r,$$

for some $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_r$, $\varepsilon_1', \varepsilon_2', \dots, \varepsilon_r'$ are elements of $\{1, -1, i, -i\}$, and the tuples $(\varepsilon_1, \varepsilon_2, \dots, \varepsilon_r)$ and $(\varepsilon_1', \varepsilon_2', \dots, \varepsilon_r')$ are not equal.

(d) Prove that there exists a nonzero polynomial P of degree at most $r - 1$, which vanishes at q, and has coefficients in $\{0, 2, 1 + i\}$ up to a factor of $\pm 1, \pm i$.

(e) Further, prove that there exists a nonzero polynomial Q of degree at most $r - 1$, which vanishes at q, and has coefficients in $\{0, 1, 1 + i\}$ up to a factor of $\pm 1, \pm i$.

(f) Write $q = \frac{a}{b}$ where a, b are relatively prime positive integers, and show that b divides the leading coefficient of Q in $\mathbb{Z}[i]$, and a divides the constant term of Q in $\mathbb{Z}[i]$, where

$$\mathbb{Z}[i] = \{x + yi : x, y \in \mathbb{Z}\}$$

is the ring of Gaussian integers.

(g) Prove further that a divides the constant term of Q in $\mathbb{Z}[i]$.

(h) Conclude that $q = 1$ is the only possible value of q.

Solution 12. Let (a, b) denote the coordinates of the point X. Let (x_n, y_n) and (x'_n, y'_n) be the coordinates of the two ants after n minutes. We have $x_0 = x'_0 = a$ and $y_0 = y'_0 = b$. Then for each n, note that $(x_n, y_n) - (x_{n-1}, y_{n-1})$ is equal to one of $(0, q^n), (q^n, 0), (0, -q^n), (-q^n, 0)$. Similarly, $(x'_n, y'_n) - (x'_{n-1}, y'_{n-1})$ is equal to one of $(0, q^n), (q^n, 0), (0, -q^n), (-q^n, 0)$. Let r be a positive integer such that $(x_r, y_r) = (x'_r, y'_r)$. Then we have

$$(a, b) + \sum_{n=1}^{r} (x_n - x_{n-1}, y_n - y_{n-1}) = (a, b) + \sum_{n=1}^{r} (x'_n - x'_{n-1}, y'_n - y'_{n-1}),$$

which implies that

$$\sum_{n=1}^{r} (x_n - x_{n-1}, y_n - y_{n-1}) = \sum_{n=1}^{r} (x'_n - x'_{n-1}, y'_n - y'_{n-1}).$$

This shows that

$$\sum_{n=1}^{r} ((x_n + iy_n) - (x_{n-1} + iy_{n-1})) = \sum_{n=1}^{r} \left((x'_n + iy'_n) - (x'_{n-1} + iy'_{n-1})\right).$$

For $1 \le n \le r$, write

$$(x_n + iy_n) - (x_{n-1} + iy_{n-1}) = q^n i^{k_n}$$

and

$$(x'_n + iy'_n) - (x'_{n-1} + iy'_{n-1}) = q^n i^{k'_n},$$

where k_n and k'_n are integers. Also observe that k_n and k'_n are not congruent modulo 4 for some n with $1 \le n \le r$, since the two ants do not take the same route within r minutes. Then we have

$$\sum_{n=1}^{r} q^n i^{k_n} = \sum_{n=1}^{r} q^n i^{k'_n},$$

which implies that

$$\sum_{n=1}^{r} q^{n-1} (i^{k_n} - i^{k'_n}) = 0.$$

Observing

$$1 - i = -i(1 + i),$$

it follows that for each n with $1 \le n \le r$, the difference $i^{k_n} - i^{k'_n}$ is equal to $0, 2, 1 + i$ up to a factor of $\pm 1, \pm i$. Hence, there exists a nonzero polynomial P of degree at most $r - 1$, which vanishes at q, and has coefficients in $\{0, 2, 1 + i\}$

up to a factor of $\pm 1, \pm i$. Since q is nonzero, it follows that P is a nonconstant polynomial, and hence, we may (and do) assume that P has a nonzero constant term. Further, dividing the coefficients of P by $1+i$, we may (and do) assume that P has coefficients in $\{0, 1, 1+i\}$ up to a factor of $\pm 1, \pm i$. Write $q = \frac{a}{b}$ for some relatively prime positive integers a, b. Considering the real and imaginary parts of the coefficients of P separately, and applying the rational root theorem to the resulting polynomials with integer coefficients, it follows that a divides the constant term of P in $\mathbb{Z}[i]$, and b divides the leading coefficient of P in $\mathbb{Z}[i]$. This implies that the square of the absolute value of a divides the square of the absolute value of the constant term of P, and the square of the absolute value of b divides the square of the absolute value of the leading coefficient of P. It follows that the square of the absolute value of each of a and b is at most 2, and hence $a = b = 1$. This shows that $q = 1$ is the only possible value of q.

Finally, we verify that $q = 1$ indeed works. Let the first ant walk north, east in the first two minutes, and the second ant walk east, north in the first two minutes. Then they are at the same point after two minutes, but have not taken the same route within that time. ■

Exercise 14 (Brazil National Olympiad 2018 Level 2 P5,). Initially, the numbers $1, 2, 3, \ldots, 10$ are written on a board. An operation consists of choosing two numbers a and b from the board, deleting them, and writing the number $a+b+\frac{ab}{f(a,b)}$, where $f(a,b)$ denotes the sum of all numbers currently on the board excluding a and b. This process is repeated until only two numbers x and y remain, with $x \geq y$. What is the maximum possible value of x?

> **Walkthrough** — Show that the sum of the product of the pairs of the numbers written on the board is an invariant.

Solution 13. Let S denote the sum of the product of the pairs of the numbers, present on the board, prior to an operation. Let a and b be the numbers chosen for the operation. Let s denote the sum of the numbers prior to the operation, excluding a and b, and let T denote the sum of the product of the pairs of the numbers on the board, after the operation. Note that

$$\begin{aligned} T &= S - ab - as - bs + \left(a + b + \frac{ab}{s}\right) s \\ &= S. \end{aligned}$$

This shows that the sum of the product of the pairs of the numbers on the board is an invariant. Initially, the numbers $1, 2, 3, \ldots, 10$ are on the board, and thus

$$S = \sum_{1 \leq i < j \leq 10} ij$$

$$= \frac{1}{2}\left((1+2+\cdots+10)^2 - (1^2+2^2+\cdots+10^2)\right)$$
$$= \frac{1}{2}\left(55^2 - 385\right)$$
$$= 1320.$$

This shows that the sum of the product of the pairs of the numbers on the board is always 1320. Finally, when only two numbers x and y remain, we have $xy = 1320$. This implies that $x \leq 1320$.

Now we show that the maximum possible value of x is 1320, that is, we show that it is possible to perform the operations in such a way that the last two numbers on the board are 1320 and 1. Indeed, if we exclude the number 1 in the first eight operations, then the last two numbers on the board are 1320 and 1. This shows that the maximum possible value of x is 1320. ■

Exercise 15 (Brazil National Olympiad 2022 Level 3 P1,). A single player game has the following rules: initially, there are 10 piles of stones with $1, 2, \ldots, 10$ stones, respectively. A movement consists of making one of the following operations:

(i) to choose 2 piles, both of them with at least 2 stones, combine them and then add 2 stones to the new pile;

(ii) to choose a pile with at least 4 stones, remove 2 stones from it, and then split it into two piles with amount of stones in those two piles to be chosen by the player.

The game continues until it is not possible to make an operation. Show that the number of piles with one stone in the end of the game is always the same, no matter how the movements are made.

Walkthrough —

(a) Let p denote the number of piles and s denote the total number of stones at some stage of the game.

(b) Note that after applying operation (i), (p, s) changes to $(p-1, s+2)$, and after applying operation (ii), (p, s) changes to $(p+1, s-2)$.

(c) Construct a polynomial in p and s using the above observations, which remains invariant under both the operations.

Solution 14. Let p denote the number of piles and s denote the total number of stones at some stage of the game. Note that after applying operation (i), we have $p-1$ piles and $s+2$ stones. Moreover, after applying operation (ii), we have $p+1$ piles and $s-2$ stones. This shows that $2p+s$ remains invariant under both the operations. Let P denote the number of piles when the game has come to an end by performing the operations in a certain sequence. Note

that there are at least $P-1$ piles containing only one stone. Let S denote the number of stones in the remaining pile. Observe that $1 \leq S \leq 3$. The total number of stones is equal to $S+(P-1)$. This yields

$$2P+S+P-1 = 2\times 10+(1+2+\cdots+10),$$

which implies

$$3P+S = 2\times 10+(1+2+\cdots+10)+1 = 20+55+1 = 76.$$

This implies that $P=25$ and $S=1$, so there are 25 piles with one stone in the end of the game. ■

> **Remark.** Note that applying the second operation to the pile obtained by applying the first operation, may yield the piles, to which the first operation was applied. Thus the game need not terminate. This problem seems to ask that the number of piles with one stone in the end of the game (when an end state is reached) is always the same, no matter how the movements are made.

Exercise 16 (Belarus National Olympiad 2022 Grade 9 Day 1 P4,). The numbers $1, 2, \ldots, 50$ are written on the board. Anya does the following operation: removes the numbers a and b from the board and writes their sum $a+b$, after which she also notes down the number $ab(a+b)$. After 49 of these operations, only one number is left on the board. Anya summed up all the 49 numbers in her notes and got S.

(a) Prove that S does not depend on the order of Anya's actions.

(b) Calculate S.

> **Walkthrough** —
>
> **(a)** Let us denote the sum of the cubes of the numbers on the board at a moment as B, and the sum of the numbers in Anya's notes at the same moment as N.
>
> **(b)** Prove that $B-3N$ is invariant under Anya's operation.
>
> **(c)** Calculate S using the invariance of $B-3N$.

Solution 15. Let B denote the sum of the cubes of the numbers on the board at a moment, and N denote the sum of the numbers in Anya's notes at the same moment. We will show that $B-3N$ is invariant under Anya's operation. Let a and b be the numbers removed from the board. Note that B changes to $B-a^3-b^3+(a+b)^3$ and N changes to $N+ab(a+b)$. This shows that $B-3N$ changes to

$$B-a^3-b^3+(a+b)^3-3(N+ab(a+b)) = B-3N.$$

Since $B - 3N$ is invariant, we have

$$1^3 + 2^3 + \cdots + 50^3 = (1 + 2 + \cdots + 50)^3 - 3S,$$

which gives

$$\begin{aligned} S &= \frac{(1 + 2 + \cdots + 50)^3 - (1^3 + 2^3 + \cdots + 50^3)}{3} \\ &= \frac{(1 + 2 + \cdots + 50)^3 - (1 + 2 + \cdots + 50)^2}{3} \\ &= \frac{1}{3}\left(\left(\frac{50 \cdot 51}{2}\right)^2 \cdot \left(\frac{50 \cdot 51}{2} - 1\right)\right) \\ &= \frac{1275^2 \cdot 1274}{3}. \end{aligned}$$

■

Exercise 17 (British Mathematical Olympiad Round 2 2023 P2)**.** For an integer $n > 1$, the numbers $1, 2, 3, \ldots, n$ are written in order on a blackboard. The following *moves* are possible:

(i) Take three adjacent numbers x, y, z whose sum is a multiple of 3 and replace them with y, z, x.

(ii) Take two adjacent numbers x, y whose difference is a multiple of 3 and replace them with y, x.

For example we could take: $1, 2, 3, 4 \xrightarrow{\text{(i)}} 2, 3, 1, 4 \xrightarrow{\text{(ii)}} 2, 3, 4, 1$. Find all n such that the initial list can be transformed into $n, 1, 2, \ldots, n-1$ after a finite number of moves.

Walkthrough —

(a)

Solution 16. Let $x_1, x_2, \ldots, x_n$ denote the numbers on the blackboard, in the order they are written, prior to a move. We will show that the modulo 3 congruence class of the following quantity is invariant under the moves:

$$S = \sum_{i=1}^{n} i x_i.$$

For move (i), we have $x_i = x$, $x_{i+1} = y$ and $x_{i+2} = z$ for some $1 \leq i \leq n - 2$. Then the change in S is

$$(iy + (i+1)z + (i+2)x) - (ix + (i+1)y + (i+2)z)$$

$$\begin{aligned}
&= -y - z + 2x \\
&\equiv -(x + y + z) \pmod 3 \\
&\equiv 0 \pmod 3.
\end{aligned}$$

For move (ii), we have $x_i = x$ and $x_{i+1} = y$ for some $1 \leq i \leq n-1$. Then the change in S is

$$iy + (i+1)x - (ix + (i+1)y) = x - y \equiv 0 \pmod 3.$$

Thus S is invariant modulo 3. Initially, we have

$$\begin{aligned}
S &= 1 \cdot 1 + 2 \cdot 2 + \cdots + n \cdot n \\
&= 1^2 + 2^2 + \cdots + n^2.
\end{aligned}$$

Let $n > 1$ be an integer such that the initial list can be transformed into $n, 1, 2, \ldots, n-1$ after a finite number of moves. Then we also have

$$\begin{aligned}
S &= 1 \cdot n + 2 \cdot 1 + 3 \cdot 2 + \cdots + n \cdot (n-1) \\
&= n + (2^2 - 2) + (3^2 - 3) + \cdots + (n^2 - n) \\
&\equiv 1^2 + 2^2 + \cdots + n^2 - (1 + 2 + \cdots + n) + n.
\end{aligned}$$

Since S is invariant modulo 3, we have

$$1^2 + 2^2 + \cdots + n^2 \equiv 1^2 + 2^2 + \cdots + n^2 - (1 + 2 + \cdots + n) + n \pmod 3,$$

which simplifies to

$$1 + 2 + \cdots + n \equiv n \pmod 3.$$

This gives

$$\frac{n(n-1)}{2} \equiv 0 \pmod 3,$$

and hence, either $n \equiv 0 \pmod 3$ or $n \equiv 1 \pmod 3$.

Claim — Let $n > 1$ be an interger satisfying $n \equiv 0 \pmod 3$ or $n \equiv 1 \pmod 3$. Then the initial list can be transformed into $n, 1, 2, \ldots, n-1$ after a finite number of moves.

Proof of the claim. Note that the move (i) can be applied to $1, 2, 3$ to get $2, 3, 1$, and the move (i) can be applied again to $2, 3, 1$ to get $3, 1, 2$. Thus we can transform $1, 2, 3$ into $3, 1, 2$. Similarly, the move (i) can be applied to $1, 2, 3, 4$ to get $1, 3, 4, 2$, and the move (i) can be applied again to $1, 3, 4, 2$ to obtain $1, 4, 2, 3$. Applying the move (ii) to $1, 4, 2, 3$ gives $4, 1, 2, 3$. Thus we can transform $1, 2, 3, 4$ into $4, 1, 2, 3$. Let k be a positive integer such that the list $1, 2, \ldots, n$ can be transformed into $n, 1, 2, \ldots, n-1$ for all $1 < n \leq 3k + 1$ such that 3 divides $n(n-1)$.

Consider the list $1, 2, \ldots, 3k+3$. Since $4, 5, \ldots, 3k+3$ are congruent to $1, 2, \ldots, 3k$ modulo 3 respectively, by the induction hypothesis, we can transform $4, 5, \ldots, 3k+3$ into $3k+3, 4, 5, \ldots, 3k+2$. Applying the move (ii) to $1, 2, 3, 3k+3, 4, 5, \ldots, 3k+2$ gives $1, 2, 3k+3, 3, 4, 5, \ldots, 3k+2$. Applying the move (i) to $1, 2, 3k+3$ twice gives $3k+3, 1, 2$. Thus we can transform $1, 2, \ldots, 3k+3$ into $3k+3, 1, 2, 3, 4, \ldots, 3k+2$.

Consider the list $1, 2, \ldots, 3k+4$. Since $4, 5, \ldots, 3k+4$ are congruent to $1, 2, \ldots, 3k+1$ modulo 3 respectively, by the induction hypothesis, we can transform $4, 5, \ldots, 3k+4$ into $3k+4, 4, 5, \ldots, 3k+3$. Applying move (i) to $1, 2, 3, 3k+4$ gives $1, 2, 3k+4, 3$. Applying move (i) to $1, 2, 3k+4$ twice gives $3k+4, 1, 2$. Thus we can transform $1, 2, \ldots, 3k+4$ into $3k+4, 1, 2, 3, 4, \ldots, 3k+3$.

This completes the induction step, and hence the claim holds by induction. □

■

Exercise 18 (Chip-firing game)**.** Assume that on the Cartesian plane, four chips are placed at the origin. In each step, you may choose a lattice point (x, y) having at least a chip on it, remove one chip from (x, y), and place one chip each at $(x+1, y)$ and one at $(x, y+1)$. Show that after a finite number of steps, there are at least two chips at some lattice point.

Remark. The problem seems to ask that after any number of finitely many moves, there will exist two coins placed at the same point.

Walkthrough —

(a)

Solution 17. Let S denote the sum of $n_{(x,y)} r^{x+y}$ over all points (x, y), where there is a chip at (x, y) at some stage of the process, and $n_{(x,y)}$ denotes the number of chips at (x, y) at that stage, and r is a real number to be determined later. If we remove a chip at (x, y) and place one chip each at $(x+1, y)$ and $(x, y+1)$, then S changes to

$$S - r^{x+y} + r^{x+1+y} + r^{x+y+1} = S + r^{x+y}(2r - 1).$$

If we choose $r = \frac{1}{2}$, then S does not change after any move. Initially, $S = 4$ since there are four chips at the origin.

If there are no two chips at the same point at some stage of the process after finitely many moves, then S is the sum of r^{x+y} over all points (x, y) where there is a chip, and this implies that

$$4 = S$$

$$\begin{aligned}
&< \sum_{(x,y)\in\mathbb{Z}^2_{\geq 0}} r^{x+y} \\
&= \sum_{x=0}^{\infty}\sum_{y=0}^{\infty} r^{x+y} \\
&= \left(\sum_{x=0}^{\infty} r^x\right)^2 \\
&= \frac{1}{(1-r)^2} \\
&= 4,
\end{aligned}$$

which is a contradiction. Thus, there are at least two chips at some lattice point after a finite number of steps. ■

Exercise 19 (Serbia IMO TST 2024 Day 1 P1,). Three coins are placed at the origin of a Cartesian coordinate system. On one move one removes a coin placed at some position (x, y) and places three new coins at $(x + 1, y)$, $(x, y + 1)$ and $(x + 1, y + 1)$. Prove that after finitely many moves, there will exist two coins placed at the same point.

Remark. The problem seems to ask that after any number of finitely many moves, there will exist two coins placed at the same point.

Walkthrough —

(a)

Solution 18. Let S denote the sum of $n_{(x,y)}r^{x+y}$ over all points (x, y), where there is a coin at (x, y) at some stage of the process, and $n_{(x,y)}$ denotes the number of coins at (x, y) at that stage, and r is a real number to be determined later. If we remove a coin at (x, y) and place three coins at $(x+1, y)$, $(x, y+1)$ and $(x+1, y+1)$, then S changes to

$$S - r^{x+y} + r^{x+1+y} + r^{x+y+1} + r^{x+1+y+1} = S + r^{x+y}(2r + r^2 - 1).$$

If we choose r such that $2r + r^2 - 1 = 0$, then S does not change after any move. Let us take $r = \sqrt{2} - 1$. Since it is root of the equation $2r + r^2 - 1 = 0$, it follows that S is invariant under the moves. Initially, $S = 3$ since there are three coins at the origin.

If there are no two coins at the same point at some stage of the process after finitely many moves, then S is the sum of r^{x+y} over all points (x, y) where there is a coin, and this implies that

$$3 = S$$

$$
\begin{aligned}
&< \sum_{(x,y)\in\mathbb{Z}^2_{\geq 0}} r^{x+y} \\
&= \sum_{x=0}^{\infty}\sum_{y=0}^{\infty} r^{x+y} \\
&= \left(\sum_{x=0}^{\infty} r^x\right)^2 \\
&= \frac{1}{(1-r)^2} \\
&= \left(1+\frac{1}{\sqrt{2}}\right)^2 \\
&= \frac{3}{2}+\sqrt{2} \\
&< 3,
\end{aligned}
$$

which is a contradiction. This proves that after any number of finitely many moves, there will exist two coins placed at the same point. ■

Exercise 20 (St Petersburg Mathematical Olympiad 2025 Round 2 Grade 11 P1,). 100 red and 99 blue piranhas were released into the empty Quiet Pool. When a piranha wants to eat, it can eat two other piranhas. If its victims are of the same color, the piranha changes its color, but if they are multicolored, it does not change. In the end, only one piranha remains. What color is it?

Walkthrough —

(a) Show that the number of blue piranhas and the number of red piranhas differ by a multiple of 4.

(b) Show that a red piranha remains at the end of the process.

Solution 19. Let r, b denote the number of red and blue piranhas, respectively, before a piranha eats. If a red piranha eats two red piranhas, then r, b changes to $r-3, b+1$ respectively. If a red piranha eats two blue piranhas, then r, b changes to $r-1, b-1$ respectively. If a blue piranha eats two red piranhas, then r, b changes to $r-1, b-1$ respectively. If a blue piranha eats two blue piranhas, then r, b changes to $r+1, b-3$ respectively. If the victims are of different colors, then r, b changes to $r-1, b-1$ respectively, regardless of the color of the eater. If R (resp. B) denotes the number of red (resp. blue) piranhas at the end of this eating, then the integers $R-B, r-b$ differ by a multiple of 4. This shows that the difference between the number of red and blue piranhas remains the same modulo 4 throughout the process. Since initially there are 100 red and 99 blue piranhas, the difference is $100-99=1$. Hence, the difference between

the number of red and blue piranhas at the end is also 1 (mod 4). Therefore, the last remaining piranha must be red, since if it were blue, the difference would be -1, which is not congruent to 1 (mod 4). ■

Remark. We can ask to determine the colors of the remaining piranhas at the end, if initially there were 100 red and 100 blue piranhas.

Exercise 21 (Dutch Mathematical Olympiad 2025 P3,). Amber has a square consisting of 10×10 square boxes. One at a time, she randomly chooses one of the hundred squares and fills it with a number. The number she puts in a square is always equal to the total number of squares already filled in that square's row and column. So in the first square she chooses she writes a 0, in the second square a 0 or a 1, depending on whether it is in a different row and a different column or not, et cetera. So in each square there will be a number from 0 to 18 inclusive. After Amber has written a number in each square, she adds up all the numbers. What is/are the possible result(s) of this addition?

Walkthrough —

(a) In a chosen square, instead of writing the sum of the number of filled squares in the row and column, let us write down the pair of numbers (r, c) where r is the number of filled squares in that row before filling the square, and c is the number of filled squares in that column before filling the square.

(b) It amounts to find the sum of the numbers of the form $r + c$ across all the squares.

(c) Try to sum up the r values across all the squares, and then sum up the c values across all the squares.

(d) Show that the result of the addition is always the same, regardless of the order in which Amber fills the squares.

Solution 20. Let us allow Bob to play along with Amber. As Amber fills a square, Bob fills the same square with a pair (r, c) where r is the number of filled squares in that row before Amber fills it, and c is the number of filled squares in that column before Amber fills it. Note that the number Amber fills in that square is $r + c$. After Amber fills all the squares, Bob will have filled all the squares with pairs of numbers. Let us denote the sum of all the numbers Amber filled in as S. We have

$$S = \sum_{(r,c)} (r + c) = \sum_{(r,c)} r + \sum_{(r,c)} c.$$

Now, let us look at the first sum $\sum_{(r,c)} r$. For each row, the value of r across the squares in that row is $0, 1, 2, \ldots, 9$ in some order. Therefore, the sum of r

across the squares in that row is $0+1+2+\cdots+9=45$. Since there are 10 rows, we have

$$\sum_{(r,c)} r = 10 \cdot 45 = 450.$$

By a similar argument, we have

$$\sum_{(r,c)} c = 10 \cdot 45 = 450.$$

Therefore, we have

$$S = \sum_{(r,c)} r + \sum_{(r,c)} c = 450 + 450 = 900.$$

Thus, the only possible result of the addition is 900. ■

Exercise 22 (RMO 2016d P4,). A box contains 4032 answer scripts out of which exactly half have odd number of marks. We choose 2 scripts randomly and, if the scores on both of them are odd number, we add one mark to one of them, put the script back in the box and keep the other script outside. If both scripts have even scores, we put back one of the scripts and keep the other outside. If there is one script with even score and the other with odd score, we put back the script with the odd score and keep the other script outside. After following this procedure a number of times, there are 3 scripts left among which there is at least one script each with odd and even scores. Find, with proof, the number of scripts with odd scores among the three left.

Walkthrough — What happens to the number of scripts with odd scores?

Solution 21. Note that under this process, the number of scripts with odd scores remains unchanged or decreases by two. Consequently, if this process is repeated few times, then the number of scripts with odd scores would remain unchanged or decrease by an even number. Note that the box contains $4032 \times \frac{1}{2} = 2016$ scripts with odd scores. So after following the process certain number of times, the number of scripts with odd number of marks will always be an even number. The number of scripts with odd scores among the three left being at least one, is equal two. ■

Exercise 23. Consider the sequence of integers

$$1, 1, 2, 3, 5, 8, 13, 21, 34, \ldots,$$

where the first two terms are 1, and each subsequent term is the sum of the previous two terms. Prove that for any positive integer n, the greatest common divisor of the n-th and $(n+1)$-st terms is 1.

> **Walkthrough** — Show that the greatest common divisor of the n-th and $(n+1)$-st terms is an invariant of the sequence.

Solution 22. For a positive integer n, let a_n denote the n-th term of the sequence. For any positive integer k, note that

$$\begin{aligned}\gcd(a_{k+2}, a_{k+1}) &= \gcd(a_{k+1} + a_k, a_{k+1}) \\ &= \gcd(a_k, a_{k+1}).\end{aligned}$$

By the principle of mathematical induction, it follows that for any positive integer n,

$$\gcd(a_n, a_{n+1}) = \gcd(a_1, a_2) = \gcd(1, 1) = 1.$$

■

§2 Strategies

Exercise 24 (Covering a mutilated chessboard)**.** Assume that from an 8×8 chessboard, two squares of different colors are removed. Prove that the remaining 62 squares can be covered by 31 dominoes, where each domino covers exactly two adjacent squares.

> **Walkthrough** — Show that there is a trail passing through all the squares of the chessboard, passing through each square exactly once, such that the adjacent squares in the trail are of different colors.

Solution 23. Consider the closed trail as shown in Fig. 1. The trail passes through all 64 squares of the chessboard, and adjacent squares in the trail are of different colors. Removing two squares of different colors from the chessboard breaks the trail into two paths, each of which admits a covering by dominoes. Hence, the remaining 62 squares can be covered by 31 dominoes.

■

Exercise 25 (China TST 1995 Day 2 P2)**.** Alice and Bob play a game with a polynomial of degree at least 4:

$$x^{2n} + \square x^{2n-1} + \square x^{2n-2} + \cdots + \square x + 1.$$

They take turns to fill the empty boxes. If the resulting polynomial has no real root, Alice wins, otherwise, Bob wins. If Alice goes first, who has a winning strategy?

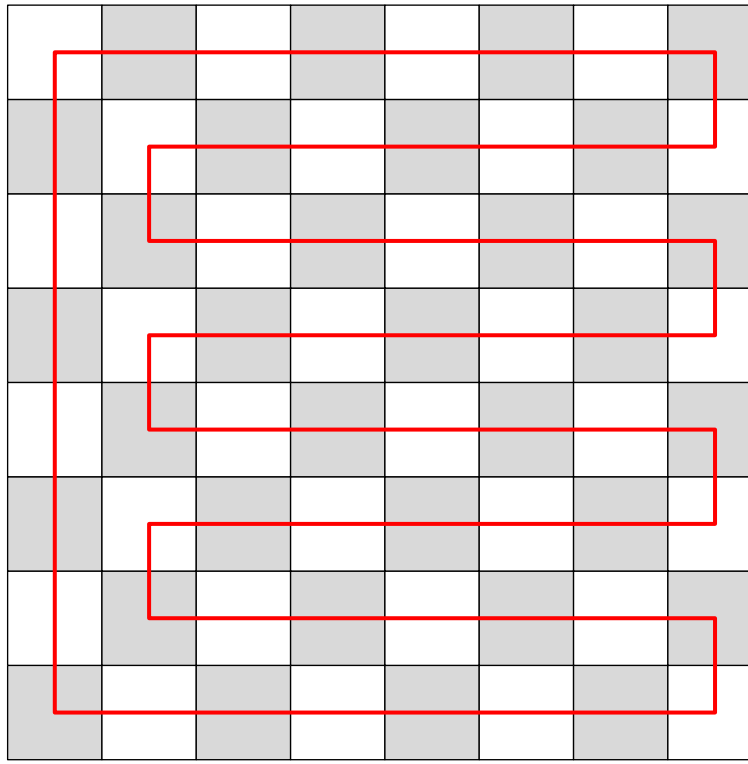


Figure 1: A closed trail through all squares of the chessboard.

Walkthrough —

(a) There are more odds than evens among the integers $1, 2, \ldots, 2n-1$.

(b) Can Bob have a winning strategy using the odds in his favour?

Solution 24. Bob has a winning strategy, as described below.

Bob makes sure that at the end of each of his turns except the last one, the number of even powers of x whose coefficients have been provided by some of them is equal to the number of odd powers of x whose coefficients have been provided by some of them. This can be done, for instance, if during a turn of Bob, other than the last turn, Bob provides the coefficient of an odd (resp. even) power of x if Alice has provided the coefficient of an even (resp. odd) power of x in the preceeding turn.

Since $n \geq 2$, it follows that Bob gets at least one turn. At the beginning of the final turn of Bob, there are two powers of x whose coefficients are to be determined, denote them by x^i, x^j, their coefficients by c_i, c_j respectively. Let $Q(x)$ denote the polynomial, obtained by the erasing the terms corresponding to x^i, x^j from the polynomial that Bob had at the beginning of his final turn. Note that

$$P(x) = Q(x) + c_i x^i + c_j x^j.$$

Note that at least of i, j is odd. Interchanging i, j if necessary, let us assume that i is odd. We descrie the strategy that Bob follows in the two cases below.

Let us consider the case that j is even. Bob determines c_j in such a way that for any choice of c_i, the completed polynomial $P(x)$ is guaranteed to have at least one real root. This can be done, for instance, by taking c_j satisfying

$$Q(1) + Q(-1) + 2c_j = 0.$$

For any choice of c_i, the above choice of c_j shows that $P(1) + P(-1) = 0$, which implies that $P(x)$ has a root in the interval $[-1, 1]$.

Let us consider the case that j is odd. Bob determines c_j in such a way that for any choice of c_i by Alice in the next turn, the completed polynomial $P(x)$ is guaranteed to have at least one real root. This can be done, for instance, by taking c_j satisfying

$$Q(2) + c_j 2^j + 2^i Q(-1) - c_j 2^i = 0.$$

Since $i \neq j$, the above holds for some $c_j \in \mathbb{R}$. For any choice of c_i, note that

$$P(2) + 2^i P(-1) = 0$$

holds, which implies that $P(x)$ has a root in $[-1, 2]$. ■

Exercise 26 (British Mathematical Olympiad Round 1 2003/4 P3)**.** Alice and Barbara play a game with a pack of $2n$ cards, on each of which is written a

positive integer. The pack is shuffled and the cards laid out in a row, with the numbers facing upwards. Alice starts, and the girls take turns to remove one card from either end of the row, until Barbara picks up the final card. Each girl's score is the sum of the numbers on her chosen cards at the end of the game. Prove that Alice can always obtain a score at least as great as Barbara's.

> **Walkthrough** —
>
> **(a)**

Solution 25. to write ■

Exercise 27 (Moldova National Olympiad 2006 Grade 11 P4, , cf. QEDMO 2006 P11)**.** On a table, there are 2006 cards, each with a natural number written on it. Two players play a game, taking turns. In each turn, a player takes a card from either the leftmost or the rightmost end of the row of cards. The game ends when all cards are taken. After the game, each player calculates the sum of the numbers on the cards they have taken. If the sum of the first player is not less than the sum of the second player, then the first player wins. Show that the first player has a winning strategy. (Note that the integers on the cards are visible to both players throughout the game.)

> **Walkthrough** —
>
> **(a)** Show that the first player can always take the cards in the odd positions or the cards in the even positions.
>
> **(b)** Before the game starts, the first player compares the sum of the numbers on the cards in the odd positions and the sum of the numbers on the cards in the even positions.

Solution 26. The first player can use the following strategy to win the game. Before the game starts, the first player calculates the sum of the numbers on the cards in the odd positions, and the sum of the numbers on the cards in the even positions. Since there are 2006 cards, one of these sums is greater than or equal to the other. If the sum of the numbers on the cards in the odd positions is greater than or equal to the sum of the numbers on the cards in the even positions, then the first player always takes the cards in the odd positions. Otherwise, the first player always takes the cards in the even positions. In either case, the first player wins.

Note that the first player can always take the cards in the odd positions or the cards in the even positions, because the first player starts the game and there are an even number of cards. Indeed, if the first player takes the card in the odd (resp. even) position in the first turn from the left (resp. right) end of the row, then after each turn of the second player, the first player can respond

by taking the card adjacent to the card taken by the second player, which is in the odd (resp. even) position. Thus, the first player can always take the cards in the odd positions or the cards in the even positions, and hence ensures that the sum of the numbers on the cards taken by the first player is greater than or equal to the sum of the numbers on the cards taken by the second player. Consequently, the first player wins. ■

Exercise 28 (Bulgaria National Olympiad 2010 Day 1 P1,). A table 2×2010 is divided to unit cells. Ivan and Peter are playing the following game. Ivan starts, and puts horizontal 1×2 domino that covers exactly two unit table cells. Then Peter puts vertical 2×1 domino that covers exactly two unit table cells. Then Ivan puts horizontal 1×2 domino. Next Peter puts vertical 2×1 domino, and so on. The player who cannot put a domino loses. Which player has a winning strategy?

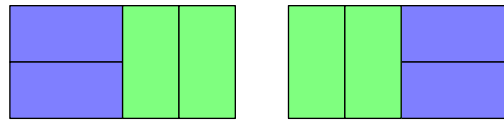
> **Walkthrough** —
>
> **(a)**

Solution 27. We claim that Ivan, the first player, has a winning strategy.

Note that the 2×2010 table can be divided into its leftmost 2×2 block, together with 502 blocks consisting of 2×4 rectangles. Denote the leftmost 2×2 block by A, and the 502 blocks by $B_1, B_2, \ldots, B_{502}$ from left to right. Then Ivan's winning strategy is as follows:

- In the first move, Ivan puts a horizontal domino in A.
- After that, if Peter puts a vertical domino in B_i for the first time for some i (for instance, a green domino as in Fig. 2a or in Fig. 2b), then Ivan puts a horizontal domino in B_i (for instance, a blue domino as in Fig. 2a or in Fig. 2b) such that another horizontal domino and a vertical domino can be put in B_i after that.
- If Peter puts a vertical domino in B_i for the second time for some i, then Ivan puts a horizontal domino in B_i.

Observe that the third step of the strategy is well-defined, since if Peter puts a vertical domino in B_i for the second time, then Ivan does have the option to put a horizontal domino in B_i by the second step of the strategy. Note that after Ivan's first move, Peter cannot put a vertical domino in A. Also, if Peter puts a vertical domino in B_i for the first time, then Ivan can put a horizontal domino in B_i following the second step of the strategy. If Peter puts a vertical domino in B_i for the second time, then Ivan can put a horizontal domino in B_i following the third step of the strategy. Thus, after every move of Peter, Ivan can follow the strategy to put a horizontal domino in some block B_i. Therefore,



(a) (b)

Figure 2: Bulgaria National Olympiad 2010 Day 1 P1, Exercise 28

Peter will run out of moves before Ivan does, and hence Ivan has a winning strategy.

■

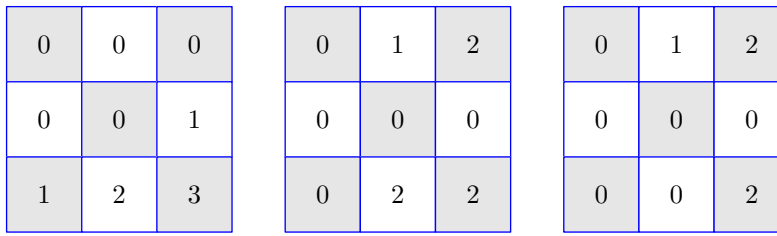
Exercise 29 (Junior Balkan MO Shortlist 2015 C1,). An $n \times n$ board (with $n \geq 3$) is divided into n^2 unit squares. Integers from 0 to n included, are written down: one integer in each unit square, in such a way that the sums of integers in each 2×2 square of the board are different. Find all n for which such boards exist.

> **Walkthrough** —
>
> **(a)**

Solution 28. Note that there are $(n-1)^2$ different 2×2 squares in the board, and the sum of the integers in each 2×2 square is at least 0 and at most $4n$. Thus, if n is such that $(n-1)^2 > 4n+1$, then by the pigeonhole principle, it would follow that there are two 2×2 squares with the same sum, and thus, no such board can exist. Hence, we must have $(n-1)^2 \leq 4n+1$, which is equivalent to $n^2 \leq 6n$. Using the fact that $n \geq 3$, we conclude that $3 \leq n \leq 6$.

Fig. 3 shows that such boards exist for $n = 3$, Fig. 4 shows that such boards exist for $n = 4$, and Fig. 5 shows that such boards exist for $n = 5$ and $n = 6$.

This proves that such boards exist precisely for $n = 3, 4, 5, 6$. ■



(a) (b) (c)

Figure 3: Junior Balkan MO Shortlist 2015 C1, Exercise 29, for $n = 3$

0	0	0	0
0	0	1	1
1	2	2	3
1	0	2	1

(a)

4	4	4	4
0	0	2	2
0	0	0	0
0	1	2	3

(b)

Figure 4: Junior Balkan MO Shortlist 2015 C1, Exercise 29, for $n = 4$

5	5	5	5	5
0	0	2	2	4
0	0	0	0	0
0	1	2	3	4
5	5	5	5	5

(a)

6	6	6	6	5	5
6	6	5	5	5	5
0	0	2	2	4	4
0	0	0	0	0	0
0	1	2	3	4	5
6	3	5	5	5	5

(b)

Figure 5: Junior Balkan MO Shortlist 2015 C1, Exercise 29, for $n = 5$ and $n = 6$

Exercise 30 (European Mathematical Cup 2016 Senior Category P2, proposed by Petar Orlić)**.** For two positive integers a and b, Ivica and Marica play the following game: Given two piles of a and b cookies, on each turn a player takes $2n$ cookies from one of the piles (with $n \geq 1$), of which he eats n and puts n of them on the other pile. Number n is arbitrary in every move. Players take turns alternatively, with Ivica going first. The player who cannot make a move, loses. Assuming both players play perfectly, determine all pairs of numbers (a, b) for which Marica has a winning strategy.

It is worth noting the apparent similarity of the above problem to Exercise 34.

Walkthrough —

(a)

Solution 29. We claim that Marica has a winning strategy if and only if $|a - b| \leq 1$.

Claim — If $|a - b| > 1$, then Ivica has a move so that the number of cookies in the resulting piles differ by at most 1. If $|a - b| \leq 1$, then any

move of Ivica results in two piles of cookies that differ by more than 1 cookie.

Proof of the claim. Let us first consider the case that $|a-b|>1$. Then there exists a nonzero integer n such that $|a-b-3n|$ lies in $\{-1,0,1\}$. If n is positive, then noting that

$$a-2n \geq b+n-1 \geq 0,$$

Ivica takes $2n$ cookies from the pile with a cookies, eats n of them and puts n on the pile with b cookies. If n is negative, then noting that

$$b+2n \geq a-n-1 \geq 0,$$

Ivica takes $-2n$ cookies from the pile with b cookies, eats $-n$ of them and puts $-n$ on the pile with a cookies. In either case, the number of cookies in the resulting piles differ by at most 1.

Now let us consider the case that $|a-b| \leq 1$. Then for any nonzero integer n, we have

$$|a-b-3n| \geq |3n| - |a-b| \geq 3-1 = 2.$$

Thus any move of Ivica results in two piles of cookies that differ by more than 1 cookie. □

Claim — Marica has a winning strategy if and only if $|a-b| \leq 1$.

Proof of the claim. If $a+b$ is equal to 1, then Marica wins. If $a=1, b=1$, then Marica wins. If a pile has two cookies and the other pile has none, then Ivica has a move which results in two piles, one with 1 cookie and the other with none, and Marica loses. This proves that if $a+b \leq 2$, then Marica has a winning strategy if and only if $|a-b| \leq 1$.

Let $n \geq 2$ be an integer such that if $a+b \leq n$, then Marica has a winning strategy if and only if $|a-b| \leq 1$.

Consider the case that $a+b = n+1$. If $|a-b| \leq 1$, then by the previous claim, any move of Ivica results in two piles of cookies that differ by more than 1 cookie, and by the induction hypothesis, Marica has a winning strategy. If $|a-b|>1$, then by the previous claim, Ivica has a move so that the number of cookies in the resulting piles differ by at most 1, and by the induction hypothesis, Ivica has a winning strategy, so Marica does not have a winning strategy.

By induction, Marica has a winning strategy if and only if $|a-b| \leq 1$. □

■

Exercise 31 (Canadian Mathematical Olympiad 2016 P1,). The integers $1, 2, 3, \ldots, 2016$ are written on a board. You can choose any two numbers on the board and replace them with their average. For example, you can replace 1

and 2 with 1.5, or you can replace 1 and 3 with a second copy of 2. After 2015 replacements of this kind, the board will have only one number left on it.

(i) Prove that there is a sequence of replacements that will make the final number equal to 2.

(ii) Prove that there is a sequence of replacements that will make the final number equal to 1000.

Walkthrough —

(a)

Solution 30.

Claim — Let x, y be positive integers with $y - x \geq 2$. There is a sequence of moves, when performed to the integers $x, x+1, \ldots, y$, results in the final number $x + 1$. Moreover, there is a sequence of moves, when performed to the integers $x, x + 1, \ldots, y$, results in the final number $y - 1$.

Proof of the claim. If $y - x = 2$, then the claim is true since we can replace x and y with their average, and next, replace the average and $x + 1$ with their average to get $x + 1$, which is equal to $y - 1$.

Assume that $t \geq 2$ be an integer such that the claim is true for all pairs of integers x, y with $y - x \leq t$. Let x, y be positive integers with $y - x = t + 1$. By the induction hypothesis, there is a sequence of moves, when performed to the integers $x+1, x+2, \ldots, y$, results in the final number $x+2$. Next, replace x and $x + 2$ with their average to get $x + 1$. Moreover, by the induction hypothesis, there is a sequence of moves, when performed to the integers $x, x+1, \ldots, y-1$, results in the final number $y - 2$. Next, replace $y - 2$ and y with their average to get $y - 1$.

By induction, the claim is true for all pairs of integers x, y with $y - x \geq 2$. □

Apply the claim with $x = 1$ and $y = 2016$ to get the final number 2.

Apply the claim with $x = 1$ and $y = 999$ to get the final number 998. Next, apply the claim with $x = 1000$ and $y = 2016$ to get the final number 1001. Finally, replace 998 and 1001 with their average to get 1000. ■

Exercise 32 (RMO 2016e P6,). A deck of 52 cards is given. There are four suites each having cards numbered $1, 2, \ldots, 13$. The audience chooses some five cards with distinct numbers written on them. The assistant of the magician comes by, looks at the five cards and turns exactly one of them face down and arranges all five cards in some order. Then the magician enters and with an agreement made beforehand with the assistant, he has to determine the face down card (both suite and number). Explain how the trick can be completed.

Solution 31. The following could be an agreement between the magician and the assistant, so that the magician can determine the face down card.

1. Given a collection of five cards, having distinct numbers written on them, the assistant puts them in a row, as follows.

2. Note that among the five cards, at least two are of the same suite.

3. The assistant chooses two cards of the same color, and puts them next to each other, and then keeps the remaining three cards to their right.

4. Note that if the integers $1, 2, \ldots, 13$ are written along the circumference of a circle Γ (in some order that is known to both the magician and the assistant, for instance, in the clockwise direction), then for any two distinct integers i, j lying between 1 and 13, one of them (call it Q) can be reached from the other one (call it P) by traversing from P along the circumference in the clockwise or in the anti-clockwise direction within at most 6 steps.

5. Denote the chosen cards of the same color, by P and Q, so that Q can be reached from P, by traversing along the circumference, in the clockwise or in the anti-clockwise direction, within at most 6 steps. The assistant puts the cards P and Q in the order Q, P (resp. P, Q) if Q can be reached from P, by traversing along the circumference, in the anti-clockwise (resp. clockwise) direction, in $s \leq 6$ steps. The assistant turns the card Q face down.

6. The assistant keeps the remaining three cards to the right of these two cards, in an order, which helps to determine the integer s as follows.

7. Denote the integers written on the remaining three cards by a, b, c with $a < b < c$. The $3! = 6$ permutations of $1, 2, 3$ can be identified with the integers $1, 2, 3, 4, 5, 6$ in some order (known to both the magician and the assistant), and the $3! = 6$ permutations of the three remaining cards can be identified with the permutations of $1, 2, 3$ in an obvious way that is known to both the magician and the assistant (for instance, sending a permutation σ of $\{a, b, c\}$ to the permutation $f \circ \sigma \circ f^{-1}$ of $\{1, 2, 3\}$ where $f : \{a, b, c\} \to \{1, 2, 3\}$ denotes the map sending a, b, c to $1, 2, 3$ respectively.). Under the composition of these two identifications, a positive integer ≤ 6 corresponds to a unique permutation of the three remaining cards. The assistant keeps the three remaining cards to the right of the first two cards, as per the permutation which corresponds to the integer s.

Since the suite of the face down card is same as the face up card which is the first in the row of the five cards, the magician determines the suite of the face down card. The magician knows that the number written on the face down

card can be reached, by traversing along Γ from the number written on the first face up card, in the clockwise (resp. anti-clockwise) direction if the face down card in the second (resp. first) one in the row of five. ■

Exercise 33 (SMMC 2017 A1,). The five sides and five diagonals of a regular pentagon are drawn on a piece of paper. Two people play a game, in which they take turns to colour one of these ten line segments. The first player colours line segments blue, while the second player colours line segments red. A player cannot colour a line segment that has already been coloured. A player wins if they are the first to create a triangle in their own colour, whose three vertices are also vertices of the regular pentagon. The game is declared a draw if all ten line segments have been coloured without a player winning. Determine whether the first player, the second player, or neither player can force a win.

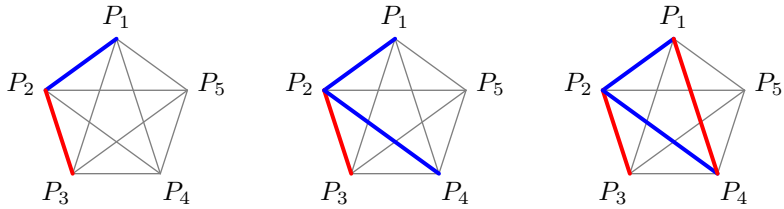


Figure 6: SMMC 2017 A1, Exercise 33, First two colored segments share a common vertex

Walkthrough — To force a win, the first player colors a side of the pentagon in blue in the first move.

Let us first consider the case that the two edges coloured in the first two moves have a common endpoint, as in Fig. 6.

(a) Show that there is no loss of generality in assuming that the first player colours P_1P_2 in blue and the second player colours P_2P_3 in red, where $P_1P_2P_3P_4P_5$ denotes the pentagon.

(b) Next, the first player colours P_2P_4 in blue.

(c) If the second player does not color P_1P_4, then in the next move the first player colours P_1P_4 in blue and wins. It remains to consider the case that the second player colours P_1P_4 in red, which we assume from now on.

(d) The first player colours P_2P_5 in blue.

(e) During the next turn of the first player, one of the edges P_1P_5 and P_4P_5 is coloured in blue, thus creating a blue triangle.

We are still left with the case that the two edges coloured in the first two moves do not have a common endpoint, as in Fig. 7.

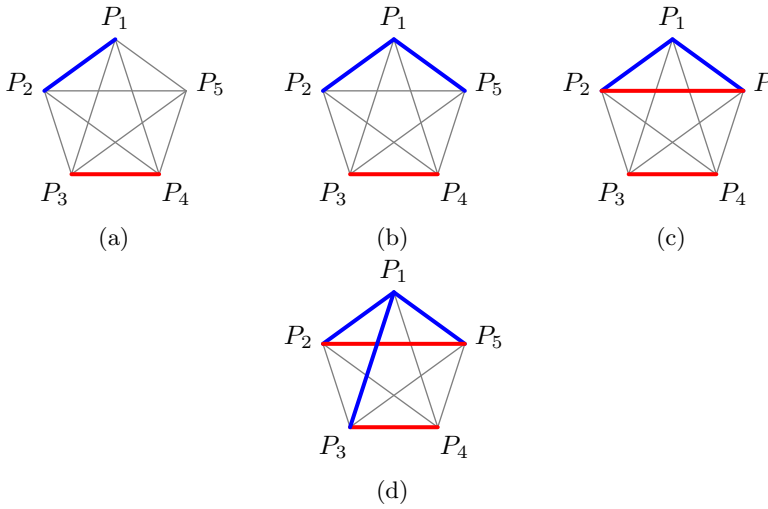


Figure 7: SMMC 2017 A1, Exercise 33, First two colored segments share no common vertex

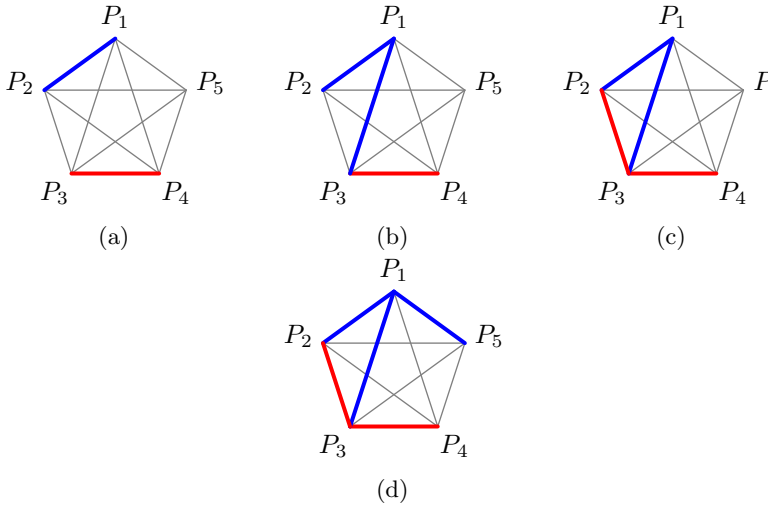


Figure 8: SMMC 2017 A1, Exercise 33, First two colored segments share no common vertex

(a) Show that there is no loss of generality in assuming that the first player colours P_1P_2 in blue and the second player colours P_3P_4 in red.

(b) Next, the first player colours P_1P_5 in blue.

(c) If the second player does not color P_2P_5, then in the next move the first player colours P_2P_5 in blue and wins. It remains to consider the case that the second player colours P_2P_5 in red, which we assume from now on.

(d) The first player colours P_1P_3 in blue.

(e) During the next turn of the first player, one of the edges P_2P_3 and P_3P_5 is coloured in blue, thus creating a blue triangle.

Here is another strategy to see that the first player can force a win, if the two edges coloured in the first two moves do not have a common endpoint, as in Fig. 8.

(a) Show that there is no loss of generality in assuming that the first player colours P_1P_2 in blue and the second player colours P_3P_4 in red.

(b) Next, the first player colours P_1P_3 in blue.

(c) This forces the second player to colour P_2P_3 in red, otherwise the first player colours P_2P_3 in blue and wins.

(d) The first player colours P_1P_5 in blue.

(e) During the next turn of the first player, one of the edges P_2P_5 and P_3P_5 is coloured in blue, thus creating a blue triangle.

Solution 32. to write ■

Exercise 34 (Balkan Mathematical Olympiad 2018 P3, , proposed by Dimitris Christophides, Cyprus)**.** Alice and Bob play the following game: They start with non-empty piles of coins. Taking turns, with Alice playing first, each player choose a pile with an even number of coins and moves half of the coins of this pile to the other pile. The game ends if a player cannot move, in which case the other player wins.

Determine all pairs (a, b) of positive integers such that if initially the two piles have a and b coins respectively, then Bob has a winning strategy.

It is worth noting the apparent similarity of the above problem to Exercise 30.

Walkthrough —

(a)

Solution 33. We prove that Bob has a winning strategy if and only if a, b are both even and have the same 2-adic valuations, and this common 2-adic valuation is even.

For a positive integer n, let $v_2(n)$ denote the largest integer k such that 2^k divides n, and it is called the 2-adic valuation of n.

Claim — If, prior to the turn of a player, the number of coins in the piles are odd, then the other player wins.

Proof of the Claim. If both piles have an odd number of coins, then the current player cannot make a move, and hence, the other player wins. □

Claim — If at some stage of the game, the number of coins in one of the piles is even and the other pile has an odd number of coins, then the game continues indefinitely, and hence, none of the players has a winning strategy.

Proof of the Claim. If at some stage of the game, one pile has an even number of coins and the other has an odd number of coins, then the current player is forced to choose the pile with an even number of coins. After moving half of the coins from the even pile to the odd pile, the resulting piles will again consist of one pile with an even number of coins and the other pile with an odd number of coins. By repeating the same argument, more precisely, by induction on the number of turns, it follows that the game continues indefinitely, and hence, none of the players has a winning strategy. □

Claim — If a, b have the same 2-adic valuations, then Bob is forced to win if and only if $v_2(a)$ is even.

Proof of the Claim. If a, b have the same 2-adic valuations, then after each turn, the resulting piles will also have the same 2-adic valuations, which is one less than the 2-adic valuations of a, b. Therefore, the game continues until both piles have an odd number of coins, and then the next player to move loses by the first claim. Note that if $v_2(a)$ is even (resp. odd), then both the piles will have an odd number of coins after an even (resp. odd) number of turns, and hence, both the piles will have an odd number of coins after some turn of Bob (resp. Alice), and consequently, Bob (resp. Alice) is forced to win. This completes the proof of the claim. □

Claim — Assume that a, b are even, and are of different 2-adic valuations. If the smaller 2-adic valuation among a, b is odd, then Alice has a winning strategy.

Proof of the Claim. Alice follows the strategy to choose the pile with the smaller 2-adic valuation in her first turn, and after her turn, the resulting piles will have the same 2-adic valuation, which is one less than the smaller 2-adic valuation among a, b, and is even. Now, Bob and Alice will alternate in taking

turns, and by the previous claim, it follows that the second player, which is now Alice, is forced to win. □

Claim — If at some stage of the game, the numbers of coins in the piles are even, and have different 2-adic valuations, and the smaller of the 2-adic valuations of the numbers of coins in the piles is even, then after the next turn, the numbers of coins in the resulting piles are even, and have different 2-adic valuations, and the smaller of the 2-adic valuations of the numbers of coins in the resulting piles is equal to the smaller of the 2-adic valuations of the numbers of coins in the piles, prior to the turn of the current player, and consequently, the game continues indefinitely, and hence, none of the players has a winning strategy.

Proof of the Claim. Let x, y be the numbers of coins in the piles at some stage of the game. Assume that x, y are even and $v_2(x) < v_2(y)$. If the current player chooses the pile with x coins, then after his/her turn, the resulting piles will have the same 2-adic valuation, which is $v_2(x)-1$, and is odd, and hence, by the previous claim, the current player is forced to lose. Therefore, the current player is forced to choose the pile with y coins. If $v_2(y) \geq v_2(x)+2$, then after the next turn, the numbers of coins in the resulting piles are even and have different 2-adic valuations, and the smaller of the 2-adic valuations of the numbers of coins in the piles is equal to $v_2(x)$, which is even. If $v_2(y) = v_2(x)+1$, then after the next turn, the numbers of coins in the resulting piles are even and have different 2-adic valuations, and the smaller of the 2-adic valuations of the numbers of coins in the piles is equal to $v_2(x)$, which is even. This proves the claim. □

This establishes that Bob has a winning strategy if and only if a, b are both even and have the same 2-adic valuations, and this common 2-adic valuation is even. ■

Exercise 35 (Mexico National Olympiad 2020 P3, , proposed by Victor and Isaías de la Fuente)**.** Let $n \geq 3$ be an integer. Two players, Ana and Beto, play the following game. Ana tags the vertices of a regular n-gon with the numbers from 1 to n, in any order she wants. Every vertex must be tagged with a different number. Then, we place a turkey in each of the n vertices. These turkeys are trained for the following. If Beto whistles, each turkey moves to the adjacent vertex with greater tag. If Beto claps, each turkey moves to the adjacent vertex with lower tag. Beto wins if, after some number of whistles and claps, he gets to move all the turkeys to the same vertex. Ana wins if she can tag the vertices so that Beto can't do this. For each $n \geq 3$, determine which player has a winning strategy.

Walkthrough —

(a)

Solution 34. If n is even, then Ana wins for any arrangement of the tags. To see this, note that the vertices of the n-gon can be colored with two colors, say red and blue, such that adjacent vertices have different colors. Since n is even, there are the same number of red and blue vertices. After a whistle or a clap, the turkeys on the red vertices will move to blue vertices, and the turkeys on the blue vertices will move to red vertices.

■

Exercise 36 (Mexico National Olympiad 2020 P4, , proposed by Victor de la Fuente)**.** Let $n \geq 3$ be an integer. In a game there are n boxes in a circular array. At the beginning, each box contains an object which can be rock, paper or scissors, in such a way that there are no two adjacent boxes with the same object, and each object appears in at least one box. Same as in the game, rock crushes scissors, scissors cut paper, and paper covers rock. The game consists of moving objects from one box to another according to the following rule:

> Two adjacent boxes and one object from each one are chosen in such a way that these are different, and we move the loser object to the box containing the winner object. For example, if we picked rock from box A and scissors from box B, we move scissors to box A.

Prove that, applying the rule enough times, it is possible to move all the objects to the same box.

Walkthrough —

(a)

Solution 35. We claim that there are three consecutive boxes such that one of them contains a rock, one of the remaining two boxes contains a scissor, and the remaining box contains a paper. Consider a box $\mathcal{B}$ containing a rock. Assume that the above claim is false. Then the box next to $\mathcal{B}$ in the clockwise direction must contain a scissor or a paper. If it contains a scissor (resp. paper), then the box next to it contains only rocks, the next box contains only scissors (resp. paper), the next box contains only rocks, and moving this way in the clockwise direction, we will arrive at a box which contains a paper (resp. a scissor), proving the claim.

Consider three boxes, containing a rock, a scissor, and a paper in some order. We mark them, and called them a marked rock, or a marked scissor,

or a marked paper. Note that each of a rock, a scissor, and a paper, wins against the stronger of the remaining two, and loses to the weaker of them. So, whichever of the first and the third box has a stronger marked element, the box in the middle capture it using its marked element. Next, the initially marked element of the middle square, moves to one of the first and the third box, which initially had a weaker marked element. Now, the only marked element from the middle square moves to the box having two marked elements. Hence, after a few moves, one can obtain a box, containing a rock, a scissor, and a paper. All the objects from the box to its left, that is, from the previous box in the anticlockwise direction, can be moved to this box. This proves that there is a box, containing at least a rock, a scissor, and a paper, so that one box to its left is empty.

Let $1 \leq k \leq n-2$ be a positive integer such that there is a box $\mathcal{C}$, containing at least a rock, a scissor, and a paper, and the k boxes to its left are empty. Let $\mathcal{D}$ denote the box to the right of the box $\mathcal{C}$, and O denote an object of $\mathcal{D}$. Note that $\mathcal{C}$ has two elements P, Q, such that O wins against the stronger of them, and loses to the weaker. Hence, the stronger of P, Q can be moved to $\mathcal{D}$, and the weaker one can be moved to $\mathcal{D}$ after that. Observe that $\mathcal{D}$ contains a rock, a scissor and a paper. Therefore, after a few moves, all objects of $\mathcal{C}$ can be moved to $\mathcal{D}$. Note that the $k+1$ boxes to the left of $\mathcal{D}$ are empty. By the principle of induction, it follows that all objects can be moved to a single box. ■

Exercise 37 (IMOSL 2020 C8, Austria)**.** Players A and B play a game on a blackboard that initially contains 2020 copies of the number 1. In every round, player A erases two numbers x and y from the blackboard, and then player B writes one of the numbers $x+y$ and $|x-y|$ on the blackboard. The game terminates as soon as, at the end of some round, one of the following holds:

1. one of the numbers on the blackboard is larger than the sum of all other numbers,
2. there are only zeros on the blackboard.

Player B must then give as many cookies to player A as there are numbers on the blackboard. Player A wants to get as many cookies as possible, whereas player B wants to give as few as possible. Determine the number of cookies that A receives if both players play optimally.

Exercise 38 (Brazil TST 2021 Test 2 Day 1 P1,)**.** Players A and B play a game on a blackboard that initially contains 2020 copies of the number 1. In every round, player A erases two numbers x and y from the blackboard, and then player B writes one of the numbers $x+y$ and $|x-y|$ on the blackboard. The game terminates as soon as, at the end of some round, one of the following holds:

1. one of the numbers on the blackboard is larger than the sum of all other numbers,
2. there are only zeros on the blackboard.

Player B has to pay to player A an amount in reais equivalent to the quantity of numbers left on the blackboard after the game ends. Show that player A can earn at least 8 reais regardless of the moves taken by B.

Gas Station Problem ISL 2021 C5

Exercise 39 (Canadian Junior Mathematical Olympiad 2022 P2,). You have an infinite stack of T-shaped tetrominoes (composed of four squares of side length 1), and an $n \times n$ board. You are allowed to place some tetrominoes on the board, possibly rotated, as long as no two tetrominoes overlap and no tetrominoes extend off the board. For which values of n can you cover the entire board?

Walkthrough —

(a)

Solution 36. Note that a 4×4 board can be covered by T-shaped tetrominoes, as shown in Fig. 9. Hence, an $n \times n$ board can be covered by T-shaped tetrominoes if n is a multiple of 4.

The following claim proves the converse.

Claim — An $n \times n$ board can be covered by T-shaped tetrominoes only if n is a multiple of 4.

Proof of the claim. Assume that an $n \times n$ board can be covered by T-shaped tetrominoes. Each T-shaped tetromino covers exactly 4 unit squares, so the total number of unit squares covered must be a multiple of 4. Since the board has n^2 unit squares, it follows that n^2 must be a multiple of 4. This implies that n itself must be a multiple of 2.

Next, we will show that n is a multiple of 4. Consider the coloring of the board in a checkerboard pattern, where the top-left square is colored black. Each T-shaped tetromino covers either three black squares and one white square, or three white squares and one black square, depending on its orientation. Therefore, each T-shaped tetromino covers an odd number of squares of each color. Since n is even, the board has an even number of black squares and an even number of white squares. To cover the entire board, we would need to cover an even number of black squares and an even number of white squares, and hence, we would need to use an even number of T-shaped tetrominoes. This implies that the total number of squares covered, which is 4 times the number of tetrominoes, must be a multiple of 8. Therefore, n^2 must be a multiple of 8, which implies that n must be a multiple of 4. □

This proves that an $n \times n$ board can be covered by T-shaped tetrominoes if and only if n is a multiple of 4. ■

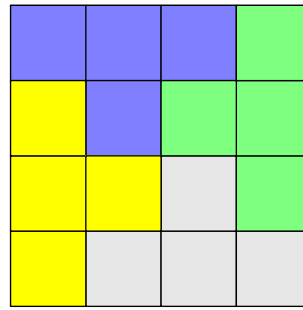


Figure 9: Canada Junior Mathematical Olympiad 2022 P2, Exercise 39

Exercise 40 (Junior Balkan MO 2023 P3, , proposed by Théo Lenoir, France)**.** Alice and Bob play the following game on a 100×100 grid, taking turns, with Alice starting first. Initially the grid is empty. At their turn, they choose an integer from 1 to 100^2 that is not written yet in any of the cells and choose an empty cell, and place it in the chosen cell. When there is no empty cell left, Alice computes the sum of the numbers in each row, and her score is the maximum of these 100 numbers. Bob computes the sum of the numbers in each column, and his score is the maximum of these 100 numbers. Alice wins if her score is greater than Bob's score, Bob wins if his score is greater than Alice's score, otherwise no one wins.

Find if one of the players has a winning strategy, and if so which player has a winning strategy.

> **Walkthrough** —
>
> **(a)**

Solution 37. Let us cover each of the 100 rows using 50 non-overlapping 1×2 dominoes. The two dominoes covering the four leftmost cells of the first row are called the *first* and *second* domino. We will show that Bob has a winning strategy, by using the following pairing strategy:

- If Alice places a number x in any cell of a domino other than the first and second domino, then Bob places $100^2 + 1 - x$ in the other cell of the same domino.
- If Alice places a number x in some cell of any of the first and second domino, then Bob places $100^2 + 1 - x$ in some cell of the other domino.

Note that Bob can always follow this strategy. Indeed, if Alice places a number x in a cell of a domino, then no cell of the grid filled earlier contains 100^2+1-x. Now, at the end of the game, the sum of the numbers in each row is $50 \cdot (100^2+1)$. On the other hand, the sum of the numbers in some column is at least $50 \cdot (100^2+1)$. Therefore, Bob's score is at least as large as Alice's score.

We will show that Bob's score is strictly greater than Alice's score. Let us assume for the sake of contradiction that Bob's score is equal to Alice's score, which is $50 \cdot (100^2+1)$. Then, the sum of the numbers in each column is $50 \cdot (100^2+1)$. In particular, the sum of the numbers in each of the first two columns is $50 \cdot (100^2+1)$. Hence, the sum of the numbers in the first domino is

$$2 \cdot 50 \cdot (100^2+1) - 99 \cdot (100^2+1) = 100^2+1.$$

Note that Bob's strategy implies that the sum of the numbers in any of the first and second domino is not 100^2+1. This is a contradiction, which completes the proof of the fact that Bob has a winning strategy. ■

> **Remark.** Dragomir Grozev argues that for the analogous problem on an $n \times n$ grid with n odd, Alice has a winning strategy.

Exercise 41 (Swiss Mathematical Olympiad 2023 Final Round Day 1 P2,). The wizard Albus and Brian are playing a game on a square of side length $2n+1$ meters surrounded by lava. In the centre of the square there sits a toad. In a turn, a wizard chooses a direction parallel to a side of the square and enchants the toad. This will cause the toad to jump d meters in the chosen direction, where d is initially equal to 1 and increases by 1 after each jump. The wizard who sends the toad into the lava loses. Albus begins and they take turns. Depending on n, determine which wizard has a winning strategy.

> **Walkthrough** —
>
> **(a)**

Solution 38. Note that if $n = 0$, then Brian wins. Henceforth, we assume that $n \geq 1$. We claim that Brian wins if he follows the strategy of sending the toad in the opposite direction of Albus's previous jump.

Assume for the sake of contradiction that Brian's strategy fails after the kth turn of Albus. Since $n \geq 1$, we have $k \geq 2$. Note that a move of Albus and the subsequent move of Brian together cause a net displacement of the toad by 1 meter. Hence, two steps prior to Brian's kth turn, that is, after Brian's $(k-1)$th turn, the toad is at the edge of the square, otherwise, after next two moves, the toad would still be in the square. Moreover, the toad reaches the edge of the square after at least $2n$ moves, since the toad starts at the centre

of the square. Therefore, $k-1 \geq n$, and hence, $k \geq n+1$. On the other hand, during k-th move of Albus, the toad jumps $2k$ meters from the edge of the square, which is more than the side length of the square, and hence, Albus loses. This is a contradiction, and hence, Brian's strategy works. ■

Exercise 42 (Columbia National Olympiad 2024 P1, , proposed by Juan David Restrepo)**.** Daniel and Juan David have three piles of stones consisting of 30 yellow stones, 20 blue stones and 10 red stones respectively. They then take turns to remove one stone from any pile, starting with Daniel. Daniel wins if at any moment there are two piles with the same positive number of stones; in any other case, Juan David wins. Determine which of the two players has a winning strategy and describe it.

Walkthrough —

(a) Show that if Daniel removes a yellow stone during his turns, then he will win.

(b) Assume that after k turns in total, only one yellow stone is left.

(c) Assume that among these k turns, Daniel has removed x yellow stones, and Juan David has removed y yellow stones, b blue stones and r red stones.

(d) Show that $b+r \leq b+r+y \leq x \leq x+y = 29$.

(e) Prove that the difference between the number of yellow stones and the number of blue stones or the difference between the number of yellow stones and the number of red stones is less than or equal to zero after k turns.

(f) Conclude that Daniel wins.

Solution 39. The first player, Daniel has a winning strategy. The strategy of Daniel is to remove a yellow stone during his turns. We claim that Daniel will win by following this strategy. More precisely, we will show that Daniel will win when only one yellow stone is left or earlier. Assume that after k turns in total, only one yellow stone is left. Assume that among these k turns, Daniel has removed x yellow stones, and Juan David has removed y yellow stones, b blue stones and r red stones. Note that after these k turns, there are $30-x-y=1$ yellow stones, $20-b$ blue stones and $10-r$ red stones. Since Daniel is the first player, it follows that at any stage of the game, the number of turns of Juan David is either equal to or one less than the number of turns of Daniel. This implies that $b+r \leq b+r+y \leq x \leq x+y = 29$. This shows that at least one of the inequalities $b \leq 19$, $r \leq 9$ holds. This gives that after k turns, at least one of the differences

$$\text{the number of yellow stones} - \text{the number of blue stones} = b-19,$$
$$\text{the number of yellow stones} - \text{the number of red stones} = r-9$$

is less than or equal to zero. Observe that these two differences were both positive at the beginning of the game. Therefore, at some moment during the game, within these k turns, one of these two differences must have become zero. In other words, at some moment during the game, one of the red and blue piles had the same positive number of stones as the yellow pile. Noting that during these k turns, the number of yellow stones is always positive, we conclude that Daniel wins. ■

Exercise 43 (Columbia Junior Mathematical Olympiad 2024 P3, , proposed by Santiago Rodriguez)**.** Let $n > 1$ be a positive integer. There are n rocks arranged in a circle, with one frog sitting on each rock. The numbers $1, 2, \ldots, n$ are written on the rocks in some order. After a signal is given, each frog looks at the number i written on the rock it occupies and then jumps i positions clockwise along the circle. Determine all values of n for which it is possible to assign the numbers $1, 2, \ldots, n$ to the rocks in such a way that, after the signal, no two frogs land on the same rock.

Walkthrough —

(a) Label the rocks as the first, second, ..., n-th rock in the clockwise order. For each $i \in \{1, 2, \ldots, n\}$, let a_i be the number written on the i-th rock.

(b) Observe that if n is a positive integer such that there is an assignment of the numbers $1, 2, \ldots, n$ to the rocks such that no two frogs land on the same rock after the signal, then the numbers $i + a_i$ for $i = 1, 2, \ldots, n$ are distinct modulo n.

(c) Show that integer n satisfying the given condition must be odd.

(d) Show that if n is odd, then there is an assignment of the numbers $1, 2, \ldots, n$ to the rocks such that no two frogs land on the same rock after the signal.

Solution 40. Let $n > 1$ be a positive integer such that there is an assignment of the numbers $1, 2, \ldots, n$ to the rocks such that no two frogs land on the same rock after the signal. Label the rocks as the first, second, ..., n-th rock in the clockwise order. For each $i \in \{1, 2, \ldots, n\}$, let a_i be the number written on the i-th rock. Then, after the signal, the frog on the i-th rock jumps to the $(i + a_i)$-th rock (where the integer $i + a_i$ is taken modulo n). Since no two frogs land on the same rock, the numbers $i + a_i$ for $i = 1, 2, \ldots, n$ are distinct modulo n. In particular, the numbers $i + a_i$ for $i = 1, 2, \ldots, n$ are congruent to the numbers $1, 2, \ldots, n$ in some order modulo n. Therefore, we have

$$(1 + a_1) + (2 + a_2) + \cdots + (n + a_n) \equiv 1 + 2 + \cdots + n \pmod{n},$$

which implies that $a_1 + a_2 + \cdots + a_n \equiv 0 \pmod{n}$. Since

$$a_1 + a_2 + \cdots + a_n = 1 + 2 + \cdots + n = \frac{n(n+1)}{2},$$

it follows that n divides $\frac{n(n+1)}{2}$. This implies that $n+1$ is even, so n is odd.

Conversely, let $n > 1$ be an odd positive integer. Consider the assignment of the numbers $1, 2, \ldots, n$ to the rocks in the order $1, 2, \ldots, n$ in the clockwise direction. Then, after the signal, the frog on the i-th rock jumps to the $(i+i)$-th rock, which is the $(2i)$-th rock modulo n. Since n is odd, the numbers $2i$ for $i = 1, 2, \ldots, n$ are distinct modulo n. Therefore, no two frogs land on the same rock after the signal.

Hence, the integers $n > 1$ for which it is possible to assign the numbers $1, 2, \ldots, n$ to the rocks in such a way that no two frogs land on the same rock after the signal are precisely the odd integers. ■

Exercise 44 (Taiwan TST 2024 Round 1 Quiz 1, , proposed by Ho-Chien Chen)**.** Let $n \geq 5$ be a positive integer. There are n stars with values 1 to n, respectively. Anya and Becky play a game. Before the game starts, Anya places the n stars in a row in whatever order she wishes. Then, starting from Becky, each player takes the left-most or right-most star in the row. After all the stars have been taken, the player with the highest total value of stars wins; if their total values are the same, then the game ends in a draw. Find all n such that Becky has a winning strategy.

> **Walkthrough** —
>
> **(a)**

Solution 41. We claim that Becky has a winning strategy if and only if $n \equiv 2 \pmod 4$. If $n \equiv 2 \pmod 4$, then compares the sum of the values of the stars in the odd positions, and the sum of the values of the stars in the even positions. Since $n \equiv 2 \pmod 4$, one of these sums is strictly greater than the other. If the sum of the values of the stars in the odd positions is greater than the sum of the values of the stars in the even positions, then Becky always takes the stars in the odd positions. Otherwise, Becky always takes the stars in the even positions. In either case, Becky wins.

If $n = 2m+1$ with $m \geq 2$, then Anya places the stars with values $1, 2, \ldots, m+1$ in the odd positions from left to right, and places the remaining stars in the even positions from left to right. Then, no matter how Becky plays, Anya takes the adjacent star, which is in an even position. Note that the sum of the values of the stars in the odd positions is

$$1 + 2 + \cdots + (m+1) = \frac{(m+1)(m+2)}{2},$$

and the sum of the values of the stars in the even positions is

$$(m+2) + (m+3) + \cdots + (2m+1) = \frac{3m}{2}(m+1).$$

Since $m \geq 2$, we have

$$\frac{3m}{2}(m+1) > \frac{(m+1)(m+2)}{2},$$

so Anya wins.

Assume that $n = 4m + 4$ with $m \geq 1$. Then Anya uses a similar strategy to force a draw or a win for Anya. First, Anya places the stars with values $m+1, m+2, \ldots, 2m$ in the first m odd positions from left to right, the stars with values $2m+1, 2m+2, \ldots, 3m$ in the next m odd positions from left to right, the stars with values $3m+1, 3m+2, \ldots, 4m$ in the first m even positions from left to right, and the stars with values $1, 2, \ldots, m$ in the next m even positions from left to right. In other words, Anya places the stars in the order

$$m+1, 3m+1, m+2, 3m+2, m+3, 3m+3, \ldots, 2m, 4m,$$
$$2m+1, 1, 2m+2, 2, \ldots, 3m, m$$

from left to right. Anya play by taking the star adjacent to the star taken by Becky in each turn.

Claim — As a consequence of the above strategy, the game ends in a draw or a win for Anya.

Proof of the Claim. Let us first consider the case when Becky takes more or an equal number of stars from the left end of the row than from the right end of the row. Note that Anya takes the stars with values $3m+1, 3m+2, \ldots, 4m$. As per Anya's strategy,

$$\begin{aligned}
&\text{the sum of the values of stars taken by Anya} \\
&\quad - \text{the sum of the values of stars taken by Becky} \\
&\geq (3m+1) + (3m+2) + \cdots + (4m) \\
&\quad - ((m+1) + (m+2) + \cdots + (2m)) \\
&\quad - |(2m+1) - 1| - |(2m+2) - 2| - \cdots - |3m - m| \\
&= 2m^2 - 2m^2 \\
&= 0.
\end{aligned}$$

Next, we consider the case when Becky takes more stars from the right end of the row than from the left end of the row. Note that Anya takes the stars with values $2m+1, 2m+2, \ldots, 3m$. As per Anya's strategy,

$$\begin{aligned}
&\text{the sum of the values of stars taken by Anya} \\
&\quad - \text{the sum of the values of stars taken by Becky} \\
&\geq (2m+1) + (2m+2) + \cdots + (3m) \\
&\quad - (1 + 2 + \cdots + m)
\end{aligned}$$

$$
\begin{aligned}
&- |(3m+1) - (m+1)| - |(3m+2) - (m+2)| - \cdots - |4m - 2m| \\
&= 2m^2 - 2m^2 \\
&= 0.
\end{aligned}
$$

This proves the claim. □

This proves that Becky has a winning strategy if and only if $n \equiv 2 \pmod 4$. ■

Exercise 45 (Mexico National Olympiad 2024 P6,). Ana and Beto play on a blackboard where all integers from 1 to 2024 (inclusive) are written. On each turn, Ana chooses three numbers a, b, c written on the board and then Beto erases them and writes one of the following numbers:

$$a + b - c, a - b + c, -a + b + c.$$

The game ends when only two numbers are left on the board and Ana cannot play. If the sum of the final numbers is a multiple of 3, Beto wins. Otherwise, Ana wins. Who has a winning strategy?

Walkthrough —

(a) Can Beto have a strategy to keep the parity of the number of integers congruent to i modulo 3 unchanged for any $i \in \{0, 1, 2\}$?

(b) Note that if Ana chooses three numbers which are congruent to $0, 1, 2$ modulo 3 in some order, then Beto will fail to do so, and otherwise Beto keep these parities unchanged.

(c) If Ana makes such a move, then Beto can write the integer among the three choices that is divisible by 3.

(d) For the next moves, can Beto follow some strategy to keep the parity of the number of integers congruent to i modulo 3 unchanged for any $i \in \{0, 1, 2\}$, and keep at least two integers divisible by 3 to ensure winning the game?

(e) Note that if Ana chooses three numbers which are congruent to any of the unordered triples $(0, 1, 2)$, $(0, 0, 1)$, $(0, 0, 2)$ modulo 3 in some order, then Beto will again fail to do so, and otherwise Beto can have a strategy as sought in the previous step.

(f) If Ana makes such a move, then does Beto have a strategy to reduce the game to a collection of an even number of integers congruent to 0 modulo 3, an odd number of integers congruent to 1 modulo 3, and an odd number of integers congruent to 2 modulo 3, similar to the initial configuration of the game, and then start afresh to win the game by induction?

Solution 42. Let us describe two rules. Next, we provide a strategy of Beto based on these two rules, and then we verify that the strategy is winning for Beto.

The first rule is that

(i) if the three numbers chosen by Ana are congruent to $0, 1, 2$ modulo 3 in some order, then Beto writes the number that is congruent to 0 modulo 3,

(ii) if the three numbers chosen by Ana are not pairwise distinct modulo 3, then Beto writes a number so that the parity of the number of integers congruent to i modulo 3 does not change for any $i \in \{0, 1, 2\}$.

The second rule is that

(i) if the three numbers chosen by Ana are congruent to $0, 1, 2$ modulo 3 in some order, then Beto writes the number that is congruent to 0 modulo 3,

(ii) if the three numbers chosen by Ana are congruent to $0, 0, 1$ (resp. $0, 0, 2$) modulo 3 in some order, then Beto writes the number that is congruent to 2 (resp. 1) modulo 3,

(iii) if the three numbers chosen by Ana are not congruent to any of the above-mentioned unordered triples (that is, to any of $(0, 1, 2)$, $(0, 0, 1)$, $(0, 0, 2)$), then Beto writes a number so that the parity of the number of integers congruent to i modulo 3 does not change for any $i \in \{0, 1, 2\}$.

The strategy of Beto is to follow the first rule in the first move. For any subsequent move, Beto follows the same rule as in the previous move if the parity of the number of integers congruent to i modulo 3 did not change in the previous move for any $i \in \{0, 1, 2\}$, and otherwise Beto follows the other rule.

Claim — The first rule is well-defined.

Proof of the claim. Note that if three integers x, y, z are congruent to $0, 1, 2$ modulo 3, then $x+y+z$ is congruent to 0 modulo 3, and using the congruences

$$\begin{aligned} x + y - z &\equiv -2z \pmod 3, \\ x - y + z &\equiv -2y \pmod 3, \\ -x + y + z &\equiv -2x \pmod 3, \end{aligned}$$

it follows that exactly one of $x + y - z, x - y + z, -x + y + z$ is congruent to 0 modulo 3.

Also note that if three integers x, y, z are not pairwise distinct modulo 3, then some two of x, y, z are congruent to each other modulo 3, and the other integer among x, y, z is congruent to at least one of $x + y - z, x - y + z, -x + y + z$ modulo 3. Therefore, Beto can write a number so that the parity of the number of integers congruent to i modulo 3 does not change for any $i \in \{0, 1, 2\}$. Hence, the first rule is well-defined. □

Claim — The second rule is well-defined.

Proof of the claim. By the first part of the previous proof, it follows that if the three integers a, b, c chosen by Ana are congruent to $0, 1, 2$ modulo 3 in some order, then there is a unique number among $a+b-c, a-b+c, -a+b+c$ that is congruent to 0 modulo 3, and hence Beto can write that number.

If the three integers a, b, c chosen by Ana are congruent to $0, 0, 1$ modulo 3 in some order, then the integers $a+b-c, a-b+c, -a+b+c$ are congruent to

$$\begin{aligned}(0+0+1)-2\times 1 &\equiv 2 \pmod 3,\\ (0+0+1)-2\times 0 &\equiv 1 \pmod 3,\\ (0+0+1)-2\times 0 &\equiv 1 \pmod 3,\end{aligned}$$

in some order, and hence Beto can write the number that is congruent to 2 modulo 3.

Similarly, if the three integers a, b, c chosen by Ana are congruent to $0, 0, 2$ modulo 3 in some order, then the integers $a+b-c, a-b+c, -a+b+c$ are congruent to

$$\begin{aligned}(0+0+2)-2\times 2 &\equiv 1 \pmod 3,\\ (0+0+2)-2\times 0 &\equiv 2 \pmod 3,\\ (0+0+2)-2\times 0 &\equiv 2 \pmod 3,\end{aligned}$$

in some order, and hence Beto can write the number that is congruent to 1 modulo 3.

Finally, if the three integers a, b, c chosen by Ana are not congruent to any of the above-mentioned unordered triples (that is, to any of $(0, 1, 2)$, $(0, 0, 1)$, $(0, 0, 2)$), then the three integers a, b, c are not pairwise distinct modulo 3, and some two of a, b, c are congruent to each other modulo 3, and the other integer among a, b, c is congruent to at least one of $a+b-c, a-b+c, -a+b+c$ modulo 3. Therefore, Beto can write a number so that the parity of the number of integers congruent to i modulo 3 does not change for any $i \in \{0, 1, 2\}$.

This proves that the second rule is well-defined. □

Claim — The strategy of Beto is winning for Beto if the game is played on any collection of at least two integers, where there are an even number of integers congruent to 0 modulo 3, an odd number of integers congruent to 1 modulo 3, and an odd number of integers congruent to 2 modulo 3.

Proof of the claim. Note that if the game is played on a collection of two integers, then the integers on the board are congruent to 1 and 2 modulo 3 in some order, or both integers on the board are congruent to 0 modulo 3, and hence, their sum is congruent to 0 modulo 3, and Beto wins. Let $k \geq 2$ be

an integer, and assume that the above strategy of Beto is winning for Beto if the game is played on any collection of at least two and at most k integers, satisfying the above-mentioned conditions on the number of integers congruent to $0, 1, 2$ modulo 3.

Suppose that the game is played on a collection of $k+1$ integers, satisfying the above-mentioned conditions on the number of integers congruent to $0, 1, 2$ modulo 3.

If Beto follows part (ii) of the first rule in the first move, then the parity of the number of integers congruent to i modulo 3 does not change for any $i \in \{0, 1, 2\}$ in the first move, and this reduces the game to a collection of at least two and at most k integers, satisfying the above-mentioned conditions on the number of integers congruent to $0, 1, 2$ modulo 3, and hence Beto wins by the induction hypothesis. Therefore, we may assume that Beto follows part (i) of the first rule in the first move.

Note that after the first move, there are an even number of integers congruent to i modulo 3 for any $i \in \{0, 1, 2\}$, and at least two integers which are congruent to 0 modulo 3. If the game ends here, then the sum of the remaining integers is congruent to 0 modulo 3, and Beto wins. It remains to consider the case that the game does not end here.

If until the end of the game, Beto follows part (iii) of the second rule, then the parity of the number of integers congruent to i modulo 3 remain even for any $i \in \{0, 1, 2\}$ throughout the game, and noting that in after an application of part (iii) of the second rule, the number of integers congruent to 0 modulo 3 is at least two, it follows that the sum of the integers left on the board at the end of the game is congruent to 0 modulo 3, and Beto wins. If prior to the end of the game, Beto follows part (i) or part (ii) of the second rule at least once, then after the first application of part (i) or part (ii) of the second rule, there are an even number of integers congruent to 0 modulo 3, and an odd number of integers congruent to 1 modulo 3, and an odd number of integers congruent to 2 modulo 3. This reduces the game to a collection of at least two and at most k integers, satisfying the above-mentioned conditions on the number of integers congruent to $0, 1, 2$ modulo 3, and hence Beto wins by the induction hypothesis. □

■

Exercise 46 (Dutch IMO TSTST 2024 P4,). Let n be a positive integer with $n \geq 3$. Consider a board of $n \times n$ boxes. In each step taken, the colors of the 5 boxes that make up the figure below change color (black boxes change to white and white boxes change to black). The figure can be rotated $90°, 180°$ or $270°$. Initially, all the boxes are white. Determine for what values of n it can be achieved, through a series of steps, that all the squares on the board are black.

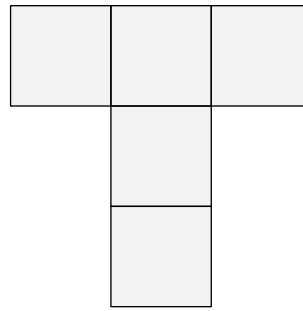


Figure 10: Dutch IMO TSTST 2024 P4, Exercise 46

> **Walkthrough** —
>
> **(a)**

Solution 43. to write ■

Exercise 47 (Austrian Mathematical Olympiad, National Competition, Preliminary Round 2025 P3, , proposed by Michael Reitmeir)**.** Consider the following game for a positive integer n. Initially, the numbers $1, 2, \ldots, n$ are written on a board. In each move, two numbers are selected such that their difference is also present on the board. This difference is then erased from the board. (For example, if the numbers $3, 6, 11$ and 17 are on the board, then 3 can be erased as $6 - 3 = 3$, or 6 as $17 - 11 = 6$, or 11 as $17 - 6 = 11$.)

For which values of n, is it possible to end with only one number remaining on the board?

> **Walkthrough** — Assume that it is possible to end with only one number remaining on the board.
>
> **(a)** Show that if d is an odd integer dividing the integers present on the board at some stage, then d divides the integer that was erased in the move prior to this stage, that is, d divides all the integers present on the board prior to this stage as well.
>
> **(b)** Show that n has no odd positive divisor other than 1.
>
> Next, assume that n is a power of 2. Use induction to show that it is possible to end with only one number remaining on the board.

Solution 44. By the claims below, it follows that it is possible to end with only one number remaining on the board if and only if n is a power of 2.

Claim — Let n be a positive integer. Assume that it is possible to end with only one number remaining on the board. Then n is a power of 2.

Proof of the Claim. Let d be an odd positive divisor of n. We will show that d must be 1. Since no two of the numbers $1, 2, \ldots, n$ differ by n, the number n cannot be erased in any move. Therefore, n must be the last remaining number on the board.

Assume that d divides all the integers present on the board at some stage in the game. Denote these integers by $a_1, a_2, \ldots, a_k$. Denote by b the integer that was erased in the move prior to this stage.

Consider the case that b is the difference of b and a_i for some i. Then b is equal to one of $b - a_i, a_i - b$. Since a_i is positive, we obtain $b = a_i - b$, which implies $a_i = 2b$. Since d divides a_i and d is odd, it follows that d divides b.

Now, consider the case that b is equal to $a_i - a_j$ for some i and j with $i \neq j$. Since d divides both a_i and a_j, it follows that d divides b.

This proves that if d divides all the integers present on the board at some stage, then d divides the integer that was erased in the move prior to this stage, that is, d divides all the integers present on the board prior to this stage as well. Since d divides n, it follows that d divides all the integers present on the board at every stage prior to the last stage. In particular, d divides 1, which implies $d = 1$.

This proves that n has no odd positive divisor other than 1. Therefore, n is a power of 2. □

Claim — Let n be a positive integer. Assume that n is a power of 2. Then it is possible to end with only one number remaining on the board.

Proof of the Claim. Write $n = 2^k$ with $k \geq 0$. Using induction, we will prove the claim. For the base case $k = 0$, we have $n = 1$. In this case, we already have only one number remaining on the board, so the claim holds. Also note that the claim holds for $k = 1$ as well, since we can erase 1 and be left with 2.

Assume that the claim holds for some $k \geq 1$, that is, for $n = 2^k$. Consider $n = 2^{k+1}$. We can divide the numbers $1, 2, \ldots, 2^{k+1}$ into two groups: $1, 2, \ldots, 2^k$ and $2^k + 1, 2^k + 2, \ldots, 2^{k+1}$. Note that

$$2^k + i = 2^{k+1} - (2^k - i)$$

for any i with $1 \leq i \leq 2^k - 1$. Therefore, we can erase the numbers $2^k + 1, 2^k + 2, \ldots, 2^{k+1} - 1$ by using the numbers $1, 2, \ldots, 2^k - 1$ and 2^{k+1}. Thus, we can reduce the numbers on the board to $1, 2, \ldots, 2^k$ and 2^{k+1}. By the induction hypothesis, we can reduce $1, 2, \ldots, 2^k$ to 2^k. Finally, when we are left with 2^k and 2^{k+1} on the board, we can erase 2^k and be left with 2^{k+1} as the only number remaining on the board. By induction, the claim holds for all $k \geq 0$. □

■

Exercise 48 (Rioplatense Mathematical Olympiad 2025 Level 1 Grade 8-9 P3,). Liko plays a game called The 24-N cards. In this game, N cards are placed on the table, each showing an integer between 1 and 9, inclusive. Some cards may have the same number. Liko wins if she can write a mathematical expression equal to 24, using each of the cards exactly once. She may only use the operations addition, subtraction, multiplication, and division, and she may group the operations in any way she wants, but multiplying by 0 is not allowed in this game. For instance, if the cards on the table are $\{7, 1, 7, 2\}$, then Liko can win by forming

$$(7 \times 7 - 1) \div 2 = 24.$$

Determine whether there exists a number N such that, regardless of what numbers appear on the N cards, Liko can always win.

> **Walkthrough** —
>
> **(a)**

Solution 45. We claim that for $N = 2 \times 68 + 9$, Liko can always win, regardless of the numbers on the cards. Given $2 \times 68 + 9$ numbers between 1 and 9, we can group them into at least 68 pairs of numbers, such that each pair consists of two equal numbers, and the remaining at most 9 numbers are pairwise distinct. These pairs of equal numbers can be used to create 68 copies of the number 1 by dividing the two equal numbers in each pair. Then, we have at least 68 copies of the number 1 and at most 8 pairwise distinct numbers between 2 and 9. Note that the sum of these at most 8 distinct numbers is at most $2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 = 44$. Using at most 44 of the 68 copies of 1, we can reduce the sum of those at most 8 distinct numbers to zero. Thus, we are left with at least $68 - 44 = 24$ copies of the number 1. Finally, we can add 24 of these 1's together to get 24. Next, we multiply 24 by the remaining numbers (which are all equal to 1), if any, to get 24 again. ■

Exercise 49 (Brazil National Olympiad 2025 Level 2 P2,). Let m be a positive integer. Ana and Banana play the following game on a 5×5 board, which is initially filled with 0's in all squares. Taking turns starting with Ana, they choose a square on the board and add an integer from the set $\{1, 2, 3, 4, 5\}$ to the number currently in that square, such that the number in the square does not exceed m. The winner is the player who makes a move that completes a row, a column, or one of the two main diagonals with the number m in all five squares. For which positive integers m does Ana, the first player, have a winning strategy?

Walkthrough —

(a)

Solution 46.

Claim — If m is a multiple of 6, then Banana has a winning strategy.

Proof of the claim. □

Claim — If m is not a multiple of 6, then Ana has a winning strategy.

Proof of the claim. □

■

Exercise 50 (Columbia National Olympiad 2025 P2, , proposed by Santiago Rodriguez)**.** The numbers $1, 2, \ldots, 2025$ are written around a circle in some order. On each turn, Celeste chooses an integer $1 \leq n \leq 2025$. Then she selects the number n and the next $n-1$ numbers following it in the clockwise direction and inverts their order on the circle. Prove that, after a finite amount of turns, Celeste can put the numbers on the circle in the order $1, 2, 3, \ldots, 2025$ in the clockwise direction, regardless of their original arrangement on the circle.

Walkthrough —

(a) Show that if the numbers $2, 3, x$ appear consecutively in that order in the clockwise direction, then after a finite number of turns, they can be arranged in the order $x, 2, 3$ in the clockwise direction, while keeping the order of the other numbers unchanged.

(b) Show that if the numbers a, b appear consecutively in that order in the clockwise direction, then after a finite number of turns, they can be arranged in the order b, a in the clockwise direction, while keeping the order of the other numbers unchanged.

(c) Using the above two results, prove that, given any arrangement of the numbers $1, 2, \ldots, 2025$ around the circle, after a finite number of turns, they can be arranged in the order $1, 2, 3, \ldots, 2025$ in this order in the clockwise direction.

Solution 47. Let us prove a few claims about the operation.

Claim — If the numbers $2, 3, x$ appear consecutively in that order in the clockwise direction, then after a finite number of turns, they can be arranged in the order $x, 2, 3$ in the clockwise direction, while keeping the

order of the other numbers unchanged.

Proof of the Claim. We can apply the operation with $n = 2$ to the three numbers $2, 3, x$ to get the arrangement $3, 2, x$. Then we apply the operation with $n = 3$ to the three numbers $3, 2, x$ to get the arrangement $x, 2, 3$. Note that the order of the other numbers is unchanged in both operations. □

Claim — Let a, b, c be three distinct integers among $1, 2, \ldots, 2025$ different from $2, 3$. If the numbers $2, 3, a, b, c$ appear consecutively in that order in the clockwise direction, then after a finite number of turns, they can be arranged in the order $2, 3, b, a, c$ in the clockwise direction, while keeping the order of the other numbers unchanged.

Proof of the Claim. We can apply the operation with $n = 3$ to the arrangement $2, 3, a, b, c$ to get the arrangement $2, b, a, 3, c$. Then we apply the operation with $n = 2$ to the arrangement $2, b, a, 3, c$ three times to get the arrangement $b, a, 3, 2, c$. Next, we apply the operation with $n = 3$ to the arrangement $b, a, 3, 2, c$ to get the arrangement $b, a, c, 2, 3$. Note that the order of the other numbers is unchanged in all operations. Applying the prior claim 2020 times, we get the arrangement $2, 3, b, a, c$, without changing the order of the other numbers. □

Claim — Let a, b, c be three distinct integers among $1, 2, \ldots, 2025$ different from $2, 3$. If the numbers a, b, c appear consecutively in this order in the clockwise direction, then after a finite number of turns, they can be arranged in the order b, a, c in the clockwise direction, while keeping the order of the other numbers unchanged.

Proof of the Claim. Applying the operation with $n = 2$ to 2 and the integer next to 2 in the clockwise direction, we obtain $2, 3$ in this order in the clockwise direction. Then we can apply the previous claim to $2, 3, a, b, c$ to get the arrangement $2, 3, b, a, c$. Next, we apply the first claim to $2, 3$ and the integer next to 3 in the clockwise direction to place $2, 3$ in this order in the clockwise direction so that 3 is placed in the same position as before. Finally, we move 2 back to its original position by applying the operation with $n = 2$ to 2 and the integer next to 2 in the clockwise direction. Note that the order of the other numbers is unchanged in all operations. □

Now, given any arrangement of the numbers $1, 2, \ldots, 2025$ around the circle, we can arrange $2, 3$ in this order in the clockwise direction. Next, applying the previous claim repeatedly, we obtain the arrangement $1, 2, 3$ in this order in the clockwise direction. Finally, we can place the remaining numbers $4, 5, \ldots, 2025$ in the correct order by repeatedly applying the previous claim to adjacent triplets of numbers that are out of order. ■

Here is an alternative solution to the problem.

Solution 48. Let us prove the following claims.

> **Claim** — If the numbers $2, 3, x$ appear consecutively in that order in the clockwise direction, then after a finite number of turns, they can be arranged in the order $x, 2, 3$ in the clockwise direction, while keeping the order of the other numbers unchanged.

Proof of the Claim. We can apply the operation with $n = 2$ to the three numbers $2, 3, x$ to get the arrangement $3, 2, x$. Then we apply the operation with $n = 3$ to the three numbers $3, 2, x$ to get the arrangement $x, 2, 3$. Note that the order of the other numbers is unchanged in both operations. □

> **Claim** — If the numbers a, b appear consecutively in that order in the clockwise direction, then after a finite number of turns, they can be arranged in the order b, a in the clockwise direction, while keeping the order of the other numbers unchanged.

Proof of the Claim. Applying the operation with $n = 2$ to 2 and the integer next to 2 in the clockwise direction, we can place $2, 3$ in this order in the clockwise direction. Then applying the prior claim to $2, 3$ and the integer next to 3 in the clockwise direction, we can obtain the arrangement $2, 3, a$ in this order in the clockwise direction. Next, applying the operation with $n = 3$ to the arrangement $2, 3, a, b$ gives the arrangement $2, b, a, 3$. Then we apply the operation with $n = 2$ to the arrangement $2, b, a, 3$ two times to get the arrangement $b, a, 2, 3$. Applying the prior claim to $2, 3$ and the integer next to 3 in the clockwise direction, we can place $2, 3$ in this order in the clockwise direction so that 3 is placed in the same position as before. Finally, we move 2 back to its original position by applying the operation with $n = 2$ to 2 and the integer next to 2 in the clockwise direction. Note that the order of the other numbers is unchanged in all operations. □

Now, given any arrangement of the numbers $1, 2, \ldots, 2025$ around the circle, we can arrange $2, 3$ in this order in the clockwise direction. Next, applying the previous claim a finite number of times, to integers lying between 3 and 4 in the clockwise direction, we can obtain the arrangement $2, 3, 4$ in this order in the clockwise direction. Next, after obtaining the arrangement $2, 3, 4, \ldots, k$ in this order in the clockwise direction, for some $4 \leq k < 2025$, we can apply the previous claim a finite number of times to integers lying between k and $k + 1$ in the clockwise direction to get the arrangement $2, 3, 4, \ldots, k, k + 1$ in this order in the clockwise direction. It follows that we can obtain the arrangement $2, 3, 4, \ldots, 2025$ in this order in the clockwise direction. This gives the desired arrangement $1, 2, 3, \ldots, 2025$ in this order in the clockwise direction. ■

Exercise 51 (All-Russian Mathematical Olympiad 2025 Grade 9 Day 2 P7, , proposed by I. A. Efremov)**.** The numbers $1, 2, 3, \ldots, 60$ are written in a row in that exact order. Igor and Ruslan take turns inserting the signs $+, -, \times$ between them, starting with Igor. Each turn consists of placing one sign. Once all signs are placed, the value of the resulting expression is computed. If the value is divisible by 3, Igor wins; otherwise, Ruslan wins. Which player has a winning strategy regardless of the opponent's moves?

Walkthrough —

(a) Igor, the first player, has a winning strategy.

(b) Note that the numbers $1, 2, 3, \ldots, 30$ are congruent to $31, 32, 33, \ldots, 60$ modulo 3, respectively.

(c) Igor in his first turn places a $-$ sign between 30 and 31.

(d) After that, the numbers $1, 2, 3, \ldots, 30$ can be thought as grouped together and the numbers $31, 32, 33, \ldots, 60$ can also be thought as grouped together.

(e) For each choice of Ruslan, Igor can respond by placing the corresponding sign in the other half.

Solution 49. We claim that Igor, the first player, has a winning strategy. Note that the numbers $1, 2, 3, \ldots, 30$ are congruent to $31, 32, 33, \ldots, 60$ modulo 3, respectively. Igor in his first turn places a $-$ sign between 30 and 31. Next, after each turn of Ruslan, which is to place a sign in one of the two halves, that is, to place a sign between two consecutive numbers between 1 and 30 or to place a sign between two consecutive numbers between 31 and 60, Igor can respond by placing the corresponding sign in the other half. Here the corresponding sign means that if Ruslan places a $\times$ sign, then Igor also places a $\times$ sign, and if Ruslan places a $+$ sign, then Igor places a $-$ sign, and if Ruslan places a $-$ sign, then Igor places a $+$ sign. After all signs are placed, the resulting expression can be written as the difference of two expressions, one of which consists of the numbers $1, 2, 3, \ldots, 30$ and the other of which consists of the numbers $31, 32, 33, \ldots, 60$. The strategy of Igor ensures that these two expressions are congruent modulo 3. Therefore, Igor wins. ■

Exercise 52 (Cono Sur Olympiad 2025 P4,)**.** Lucero and Pablo play a game on the board shown in Fig. 11. Lucero plays first, and they take turns. In each turn, a player chooses an unpainted circle from the bottom row and paints it blue, green, or red. This continues for four turns until the entire bottom row is painted.

Then, the rest of the board is painted according to the following rules:

(i) If two adjacent circles in a row are the same color, the circle above and adjacent to them is painted that same color.

(ii) If two adjacent circles in a row are of different colors, the circle above and adjacent to them is painted with the third color.

This procedure is repeated until all the circles on the board are painted. Lucero wins if the single top circle is painted red or green, and Pablo wins if it is painted blue.

Determine who has a winning strategy.

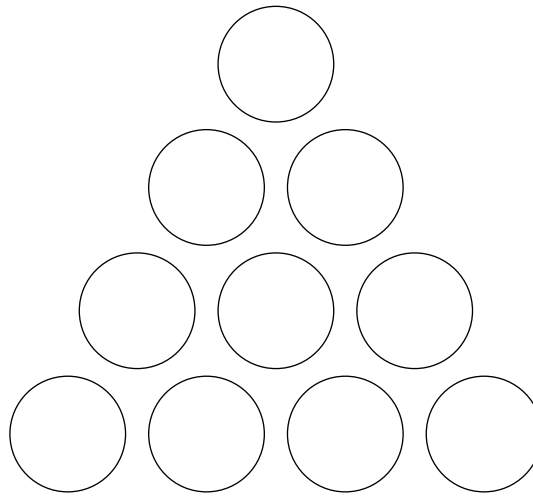


Figure 11: Cono Sur Olympiad 2025 P4, Exercise 52

Walkthrough —

(a) Use the congruence classes modulo 3 to represent the colors of the circles. For example, we can represent blue as 0, green as 1, and red as -1 modulo 3.

(b) Let a, b, c, d be the colors of the circles in the bottom row from left to right, respectively, represented as congruence classes modulo 3.

(c) Express the color of the top circle as a function of the colors of the circles in the bottom row.

(d) Determine a winning strategy for Pablo based on the colors of the circles in the bottom row.

Solution 50. We will show that Pablo has a winning strategy.

Let us consider the colors as congruence classes modulo 3 as follows: 0 for blue, 1 for green, and -1 for red. We can represent the colors of the circles in the bottom row, from left to right, as a sequence of four congruence classes, which we can denote as a, b, c, d, where each of a, b, c, d is in $\{0, 1, -1\}$. Note that if the colors of two adjacent circles are x, y modulo 3, then the color of the

circle above them is $-(x+y)$ modulo 3. Indeed, if the colors of two adjacent circles are x, y modulo 3, and x, y are congruent modulo 3, then the color of the circle above them is x modulo 3, which is $-(x+y)$ modulo 3 since $x \equiv y \pmod 3$. If x, y are not congruent modulo 3, then the color of the circle above them is the third color, which is $-(x+y)$ modulo 3. Therefore, the color of the top circle can be expressed as a function of a, b, c, d as follows. In fact, note the colors of the circle in the third row are $-(a+b)$, $-(b+c)$, and $-(c+d)$ modulo 3. Then, the colors of the circles in the second row are

$$-(-(a+b)+-(b+c)) = a+2b+c,$$
$$-(-(b+c)+-(c+d)) = b+2c+d$$

modulo 3. Finally, the color of the top circle is

$$-(a+2b+c+b+2c+d) = -a-3b-3c-d \equiv -a-d \pmod 3.$$

Therefore, the color of the top circle is determined by the colors of the leftmost and rightmost circles in the bottom row.

If Lucero, in her first turn, chooses to color the leftmost or the rightmost circle in the bottom row, then Pablo colors the circle on the opposite end of the bottom row with its negative congruence class modulo 3. This ensures that the color of the top circle is 0 modulo 3, which means that the top circle is blue, and Pablo wins.

If Lucero, in her first turn, chooses to color any of the two middle circles in the bottom row, then Pablo colors the other middle circle with any color. In next turn, Lucero colors one of the two remaining circles in the bottom row, and Pablo colors the other remaining circle in the bottom row with its negative congruence class modulo 3. This again ensures that the color of the top circle is 0 modulo 3, which means that the top circle is blue, and Pablo wins.

Therefore, Pablo has a winning strategy, and the answer is Pablo. ■

Exercise 53 (Singapore Junior Mathematical Olympiad 2025 P3,). Jack and Jill play the following game: Jack throws 3 dice and Jill can select some of them, possibly none, and turn each of them to the opposite side. Jill wins if the sum of the values on the dice is a multiple of 4. Can Jill always win? (Note the game is played with standard dice where the sum of the numbers on opposite sides is 7.)

> **Walkthrough** —
>
> **(a)**

Solution 51. Note that if the number on the top face of a die is is 1 or 2 modulo 4, that is, if it is 1, 2, 5 or 6, then for any $i \in \{1, 2\}$, the die can be flipped (if required) to make the value on the top face congruent to i modulo

4. Also if the number on the top face of a die is 3 or 0 modulo 4, that is, if it is 3 or 4, then for any $i \in \{0, 3\}$, the die can be flipped (if required) to make the value on the top face congruent to i modulo 4. Jill can always win by the following strategy:

- If the values on the top faces of all the dice lie in $\{1, 2, 5, 6\}$, then Jill can flip some of the dice to make the values on the top faces congruent to $1, 1, 2$ modulo 4.
- If the values on the top faces of exactly two dice lie in $\{1, 2, 5, 6\}$, then Jill can flip some of the dice to make the values on the top faces congruent to $2, 2, 4$ modulo 4.
- If the values on the top faces of exactly one die lie in $\{1, 2, 5, 6\}$, then Jill can flip some of the dice to make the values on the top faces congruent to $1, 3, 4$ modulo 4.
- If the values on the top faces of all the dice lie in $\{3, 4\}$, then Jill can flip some of the dice to make the values on the top faces congruent to $4, 4, 4$ modulo 4.

Using the above strategy, Jill can make the sum of the values on the top faces of the dice congruent to 0 modulo 4, and hence win the game. ■

§3 Optimization problems

Exercise 54 (Princeton University Mathematics Competition 2009 Combinatorics A6,). We have a 6×6 square, partitioned into 36 unit squares. We select some of these unit squares and draw some of their diagonals, subject to the condition that no two diagonals we draw have any common points. What is the maximal number of diagonals that we can draw?

> **Walkthrough** —
>
> **(a)**

Solution 52. to write ■

> **Remark.** See also this on AoPS.

Exercise 55 (Tournament of Towns 2014 Spring Junior A Level P2, , by A. V. Shapovalov). Peter marks several cells on a 5×5 board. Basil wins if he can cover all marked cells with three-cell corners. The corners must be inside the board and not overlap. What is the least number of cells Peter should mark to prevent Basil from winning? (Cells of the corners must coincide with the cells of the board).

Walkthrough —

(a)

Solution 53. to write ■

Exercise 56 (Tournament of Towns 2014 Spring Senior A Level P3, by I. I. Bogdanov)**.** The King called two wizards. He ordered First Wizard to write down 100 positive real numbers (not necessarily distinct) on cards without revealing them to Second Wizard. Second Wizard must correctly determine all these numbers, otherwise both wizards will lose their heads. First Wizard is allowed to provide Second Wizard with a list of distinct numbers, each of which is either one of the numbers on the cards or a sum of some of these numbers. He is not allowed to tell which numbers are on the cards and which numbers are their sums. Finally the King tears as many hairs from each wizard's beard as the number of numbers in the list given to Second Wizard. What is the minimal number of hairs each wizard should lose to stay alive?

Walkthrough —

(a)

Solution 54. to write. ■

Exercise 57 (Junior Balkan MO 2015 P4,)**.** An L-shape is one of the following four pieces, each consisting of three unit squares:

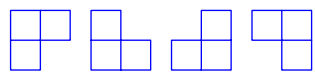


Figure 12: Junior Balkan MO 2015 P4, Exercise 57

A 5×5 board, consisting of 25 unit squares, a positive integer $k \leq 25$ and an unlimited supply of L-shapes are given. Two players A and B, play the following game: starting with A they play alternatively mark a previously unmarked unit square until they marked a total of k unit squares.

A placement of L-shapes on unmarked unit squares is called *good* if the L-shapes do not overlap and each of them covers exactly three unmarked unit squares of the board.

B wins if every good placement of L-shapes leaves uncovered at least three unmarked unit squares. Determine the minimum value of k for which B has a winning strategy.

Exercise 58 (British Mathematical Olympiad Round 1 2015/16 P4)**.** James has a red jar, a blue jar and a pile of 100 pebbles. Initially both jars are

empty. A move consists of moving a pebble from the pile into one of the jars or returning a pebble from one of the jars to the pile. The numbers of pebbles in the red and blue jars determine the state of the game. The following conditions must be satisfied:

(a) The red jar may never contain fewer pebbles than the blue jar;

(b) The game may never be returned to a previous state.

What is the maximum number of moves that James can make?

Walkthrough — Suppose there are n pebbles at the beginning, where n is even.

(a) Note that the state can be thought as a pair (r, b) where r and b denote the number of pebbles in the red and the blue jar respectively, at some instance.

(b) Observe that $r \geq b$ and $r + b \leq n$.

(c) This shows that (r, b) lies in a specific triangular region on the plane.

(d) Conclude that the maximum number of moves that James can make is at most the total number of lattice points in this triangular region, which is equal to

$$T := \sum_{r=0}^{n} \sum_{b=0}^{\min\{r, n-r\}} 1 = 1 + 3 + 5 + \cdots + (n+1) = (n/2+1)^2.$$

(e) Show that there is a path in this triangular region beginning at the origin, and traversing through at least

$$T - \frac{n}{2}$$

points in this region. Does Figs. 13 and 14 help?

(f) Observe that each move changes the parity of the sum of the coordinates.

(g) Show that after make any number of moves, the number of states with an even sum of coordinates is at most one more than the number of states with an odd sum of coordinates.

(h) Prove that the number of points in the triangular region with an even sum of coordinates is precisely $n/2 + 1$ more than those with an odd sum of coordinates. See Figs. 15 and 16.

(i) If James makes an odd number of moves, say $2k - 1$ moves, then show that the number of points on the path formed is

$$2k \leq 2O = T - n/2 - 1.$$

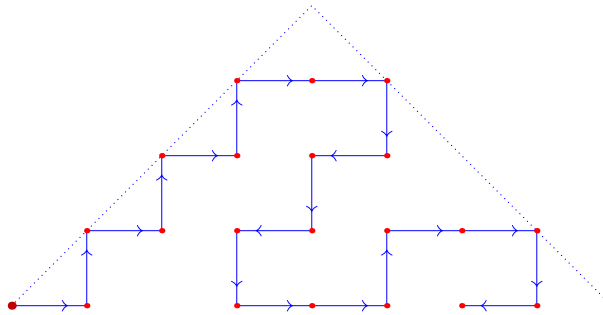


Figure 13: British Mathematical Olympiad Round 1 2015/16 P4, Exercise 58

(j) Suppose James makes an even number of moves, say, $2k$ moves. Conclude that the number of points on the path formed is

$$k + k + 1 \leq 2O + 1 = T - n/2 - 1 + 1 = T - n/2.$$

(k) Show that the maximum number of moves that James can make is

$$T - n/2 - 1 = (n/2 + 1)^2 - n/2 - 1 = \frac{n^2 + 2n}{4}.$$

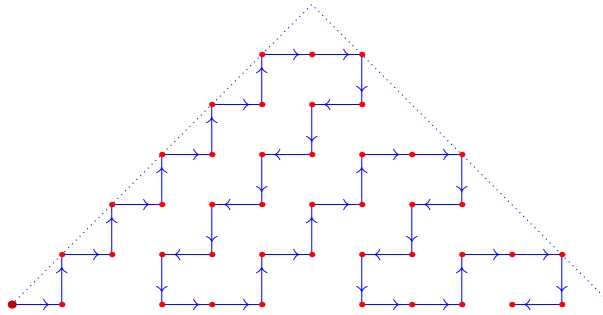


Figure 14: British Mathematical Olympiad Round 1 2015/16 P4, Exercise 58

Exercise 59 (Moscow Mathematical Olympiad 2018 Grade 11 Day 2 P2, , proposed by M. A. Evdokimov)**.** A number of dominoes were placed without overlapping on a 2018×2018 square board, each covering exactly two squares. It turned out that no two dominoes share a common whole side, that is, no two dominoes form either a 2×2 square or a 4×1 rectangle. Is it possible to cover more than 99% of all the squares on the board?

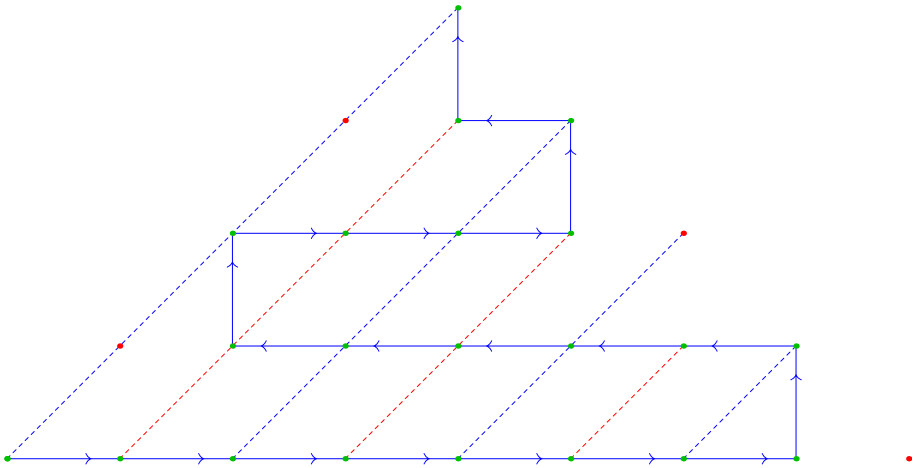


Figure 15: British Mathematical Olympiad Round 1 2015/16 P4, Exercise 58

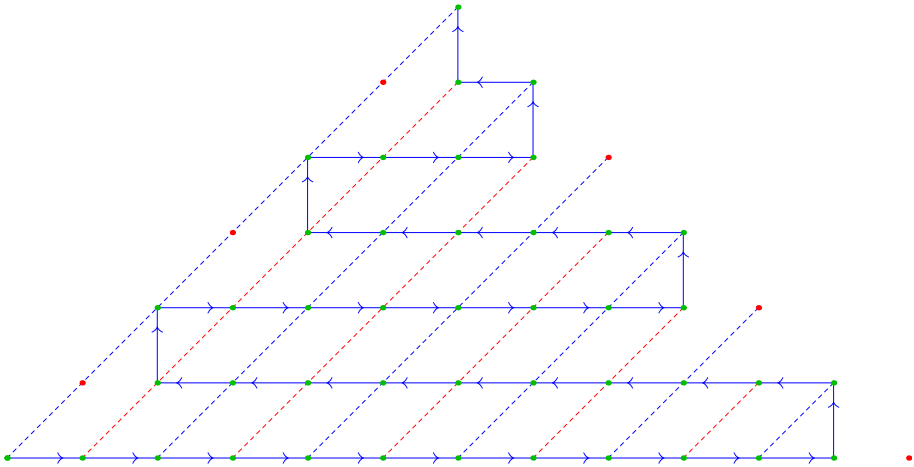


Figure 16: British Mathematical Olympiad Round 1 2015/16 P4, Exercise 58

Walkthrough —

(a)

Solution 55. to write ■

Exercise 60 (Moscow Mathematical Olympiad 2018 Grade 11 Day 2 P5, proposed by P. A. Borodin)**.** Zhenya painted a spherical egg sequentially using five different dyes, dipping it into a glass containing the current dye in such a way that exactly half of the egg's surface area (a hemisphere) was colored. As a result, the egg became completely colored. Prove that one of the dyes was superfluous, that is, if Zhenya had omitted that particular dye but had dipped the egg into the remaining dyes in the exact same manner, the egg would still have ended up completely colored.

Walkthrough —

(a)

Solution 56. to write ■

Remark. We refer to Carathéodory's theorem on convex hulls.

Exercise 61 (Benelux Mathematical Olympiad 2020 P2)**.** Let N be a positive integer. A collection of $4N^2$ unit tiles with two segments drawn on them as shown in Fig. 17 is assembled into a $2N \times 2N$ board. Tiles can be rotated.

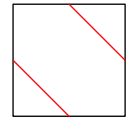


Figure 17: BxMO 2020 P2, Exercise 61

The segments on the tiles define paths on the board. Determine the least possible number and the largest possible number of such paths.

For example, there are 9 *paths on the* 4×4 *board shown in Fig. 18.*

Walkthrough —

(a)

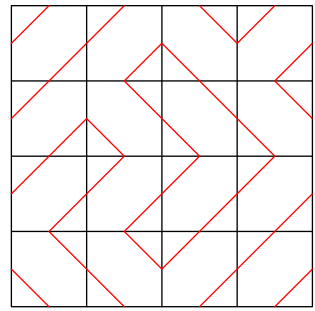


Figure 18: BxMO 2020 P2, Exercise 61

Solution 57. Consider a given tiling of a $2N \times 2N$ board using tiles of the given type. Note that there are two types of paths. Some path are between two points lying on the boundary of the square, and the other paths are contained in the interior of the square, and form cycles. Let $\mathcal{B}, \mathcal{C}$ denote the collection of paths of the first and second kind respectively. Observe that among the mid-points of the sides of these $4N^2$ tiles, there are precisely $4 \times 2N = 8N$ mid-points which lie on the boundary of the square. Since each of the paths, which are not cycles, have precisely two such points as end-points, and no two paths of these kind have an end-point in common, it follows that there are precisely $4N$ paths, which are not cycles.

Observe that no two paths have a segment in common. Note that the remaining paths, which are contained in the interior of the square, form cycles. Each of these cycles contains at least four segments. Hence, these cycles together contain at least $4|\mathcal{C}|$ segments. Also note that a path contains at least two segments, unless it is formed by single segment, and note that there are at least four such paths. Let s denote the number of paths consisting of a single segment. Since there are a total of $4N^2$ tiles, each contributing two segments, we have $8N^2$ segments in total. This shows that

$$8N^2 \geq s + 2(|\mathcal{B}| - s) + 4|\mathcal{C}|,$$

which yields

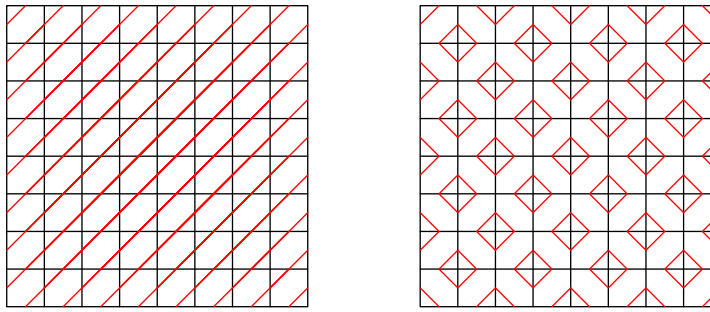
$$8N^2 + 8N \geq 4(|\mathcal{B}| + |\mathcal{C}|) - s.$$

It follows that

$$|\mathcal{B}| + |\mathcal{C}| \leq 2N^2 + 2N + \frac{s}{4} \leq 2N^2 + 2N + 1.$$

This proves that there are exactly $4N$ paths which are not cycles, and there are at most $2N^2 + 2N + 1$ paths in total.

The tiling, as illustrated in Figure 19a in the case $N = 4$, shows an example of a configuration where the number of paths is equal to $4N$. The tiling, as illustrated in Figure 19b in the case $N = 4$, shows an example of a configuration where the number of paths is equal to $2N^2 + 2N + 1$.



(a) (b)

Figure 19: BxMO 2020 P2, Exercise 61

Hence, the least possible number and the maximum possible number of paths are $4N$ and $2N^2 + 2N + 1$ respectively. ■

Exercise 62 (Swiss Mathematical Olympiad 2023 Final Round Day 2 P8,). Let n be a positive integer. We start with n piles of pebbles, each initially containing a single pebble. One can perform moves of the following form: choose two piles, take an equal number of pebbles from each pile and form a new pile out of these pebbles. Find (in terms of n) the smallest number of nonempty piles that one can obtain by performing a finite sequence of moves of this form.

> **Walkthrough** —
>
> **(a)**

Solution 58. to write ■

Exercise 63 (Kosovo National Olympiad 2024 Grade 8 P3,). The numbers $1, 2, 3, \ldots, 20$ are placed in any order in a circle. A number is considered to be *high* if it is larger than the two numbers near it. What is the greatest possible number of high numbers?

> **Walkthrough** —
>
> **(a)** Prove that there are at most 10 high numbers in any arrangement of $1, 2, \ldots, 20$ on a circle.
>
> **(b)** Find an arrangement of $1, 2, \ldots, 20$ on a circle having 10 high numbers.

Solution 59. Suppose we are given an arrangement of $1, 2, \ldots, 20$ on a circle. Let k denote the number of high numbers in this arrangement. Since 20 is a

high number, we have $k \geq 1$. Let $h_1, h_2, \ldots, h_k$ denote the high numbers in this arrangement. Assume that further that these high numbers are arranged in the order $h_1, h_2, \ldots, h_k$ on the circle. Let us consider case that $k \geq 2$. Observe that two high numbers cannot be adjacent to each other. Therefore, there must be at least one non-high number between any two high numbers. Therefore, there are at least k non-high numbers in this arrangement. Since there are 20 numbers in total, there are at most $20 - k$ non-high numbers. Therefore, we have $k \leq 20 - k$, which implies that $k \leq 10$ under the assumption that $k \geq 2$. This proves that $k \leq 10$, that is, there are at most 10 high numbers in any arrangement of $1, 2, \ldots, 20$ on a circle.

Consider the arrangement

$$1, 11, 2, 12, 3, 13, 4, 14, 5, 15, 6, 16, 7, 17, 8, 18, 9, 19, 10, 20$$

on a circle. In this arrangement, the high numbers are 11, 12, 13, 14, 15, 16, 17, 18, 19, 20. Therefore, there are 10 high numbers in this arrangement.

Therefore, the greatest possible number of high numbers is equal to 10. ■

Exercise 64 (Junior Balkan MO 2019 P4,). A 5×100 table is divided into 500 unit square cells, where n of them are coloured black and the rest are coloured white. Two unit square cells are called adjacent if they share a common side. Each of the unit square cells has at most two adjacent black unit square cells. Find the largest possible value of n.

> **Walkthrough** —
>
> **(a)**

Solution 60. to write ■

Exercise 65 (Argentina National Olympiad 2020 Level 2 P2,). Let n be a positive integer. There are n colors available. Each of the integers from 1 to 1000 must be painted with one of the n colors such that any two different numbers, if one divides the other, are painted in different colors. Determine the smallest value of n for which this is possible.

> **Walkthrough** —
>
> **(a)**

Solution 61.

■

Exercise 66 (Brazil National Olympiad 2022 Level 3 P6,). Some cells of a 10×10 chessboard are colored blue. A set of six cells is called *gremista* when the cells are the intersection of three rows and two columns, or two rows and three columns, and are painted blue. Determine the greatest value of n for which it is possible to color n chessboard cells blue such that there is not a gremista set.

> **Walkthrough** —
>
> **(a)**

Solution 62.

■

> **Remark.** This problem is equivalent to finding the maximum possible number of edges in a (possibly directed) graph on 10 vertices which is $K_{3,2}$-free.

Exercise 67 (Junior Balkan MO Shortlist 2022 C1,). Anna and Bob, with Anna starting first, alternately color the integers of the set $S = \{1, 2, \ldots, 2022\}$ red or blue. At their turn each one can color any uncolored number of S they wish with any color they wish. The game ends when all numbers of S get colored. Let N be the number of pairs (a, b), where a and b are elements of S, such that a, b have the same color, and $b - a = 3$. Anna wishes to maximize N. What is the maximum value of N that she can achieve regardless of how Bob plays?

> **Walkthrough** —
>
> **(a)**

Solution 63.

■

Exercise 68 (Junior Balkan MO Shortlist 2022 C3,). There are 200 boxes on the table. In the beginning, each of the boxes contains a positive integer (the integers are not necessarily distinct). Every minute, Alice makes one move. A move consists of the following. First, she picks a box X which contains a number c such that $c = a + b$ for some numbers a and b which are contained in some other boxes. Then she picks a positive integer $k > 1$. Finally, she removes c from X and replaces it with kc. If she cannot make any moves, she stops. Prove that no matter how Alice makes her moves, she won't be able to make infinitely many moves.

> **Walkthrough** —
> **(a)**

Solution 64.

■

Exercise 69 (Junior Balkan MO Shortlist 2023 C1,). Given is a square board with dimensions 2023×2023, in which each unit cell is colored blue or red. There are exactly 1012 rows in which the majority of cells are blue, and exactly 1012 columns in which the majority of cells are red. What is the maximal possible side length of the largest monochromatic square?

> **Walkthrough** —
> **(a)**

Solution 65.

■

Exercise 70 (Canadian Junior Mathematical Olympiad 2023 P3, , cf. CMO 2023 P1). William is thinking of an integer between 1 and 50, inclusive. Victor can choose a positive integer m and ask William: "does m divide your number?", to which William must answer truthfully. Victor continues asking these questions until he determines William's number. What is the minimum number of questions that Victor needs to guarantee this?

> **Walkthrough** —
> **(a)**

Solution 66.

■

Exercise 71 (Rioplatense Mathematical Olympiad 2025 Level 3 Grade 12 P2,). Let $n \geq 2$ be an integer. There are n numbers, each one starts at 0. In each operation, one chooses two numbers, the smallest increases by 1 and the biggest becomes 0 (if they are equal, one increases 1 and the other becomes 0). This process can be repeated indefinitely. Find, in function of n, the biggest number that can appear in the sequence.

Exercise 72 (All-Russian Mathematical Olympiad 2025 Grade 10 Day 2 P5, , also 11.5, proposed by I. I. Bogdanov). Let n be a natural number. The numbers $1, 2, \ldots, n$ are written in a row in some order. For each pair of adjacent numbers, their greatest common divisor is calculated and written on a sheet. What is the maximum possible number of distinct values among the $n - 1$ integers obtained?

Walkthrough —
(a)

Solution 67. In the following, we assume that $n \neq 1$. Using division algorithm, for any two distinct positive integers a and b, it follows that

$$\gcd(a,b) \leq \begin{cases} a \leq \frac{b}{2} & \text{if } a \text{ divides } b, \\ \frac{a}{2} & \text{if } a \text{ does not divide } b, \end{cases}$$

and hence $\gcd(a,b) \leq \frac{1}{2}\max(a,b)$. This implies that the GCD of any two adjacent numbers in the row is at most $n/2$. Therefore, the maximum possible number of distinct GCDs obtained, is at most $\lfloor n/2 \rfloor$.

We will show that this upper bound can be achieved by the following arrangement of the numbers $1, 2, \ldots, n$. For each odd positive integer $k \leq n$, write the integers between 1 and n in a block $\mathcal{B}_k$, having k as its largest odd divisor, in an increasing order, that is, the elements of $\mathcal{B}_k$ are put in the order

$$k, 2k, 4k, \ldots, 2^{e_k}k,$$

where e_k is the largest non-negative integer such that $2^{e_k}k \leq n$. Next, we put the blocks $\mathcal{B}_k$ in a row as k increases across the odd positive integers less than or equal to n.

Claim — Let $1 \leq m \leq n/2$ be a positive integer. Then m is equal to the GCD of some pair of adjacent numbers in the above arrangement.

Proof of the Claim. Since m is at most $n/2$, it follows that $2m$ is at most n. If k denotes the largest odd divisor of m, then k is less than or equal to m and hence $k \leq n$. Hence, in the above arrangement, the block $\mathcal{B}_k$ has appeared, and the numbers m and $2m$ are adjacent in this block. Note that the GCD of m and $2m$ is m. This proves the claim. □

Therefore, the GCDs of the adjacent numbers in the above arrangement include all the integers $1, 2, \ldots, \lfloor n/2 \rfloor$. Since the number of distinct GCDs obtained is at most $\lfloor n/2 \rfloor$, it follows that the maximum possible number of distinct GCDs obtained is $\lfloor n/2 \rfloor$. ■

Remark. One may also observe that if ℓ denotes the largest GCD obtained from the adjacent numbers in the row, then it is the gcd of two distinct integers between 1 and n, and the larger of these two integers is at least 2ℓ, implying that $\ell \leq n/2$.

Exercise 73 (RMO 2016g P2,). On a stormy night ten guests came to dinner party and left their shoes outside the room in order to keep the carpet clean. After the dinner there was a blackout, and the guests leaving one by one, put on at random, any pair of shoes big enough for their feet. (Each pair of shoes stays together). Any guest who could not find a pair big enough spent the night there. What is the largest number of guests who might have had to spend the night there?

Walkthrough — What happens if a person having shoe of the smallest size wears a shoe of the largest size, and next, if a person having shoe of the second smallest size wears a shoe of the second largest size, and it continues?

Solution 68. If a person having shoe of the k-th smallest size wears a shoe of the k-th largest size for $k = 1, 2, 3, 4, 5$, and if the sizes of the shoes of the ten persons are distinct, then none of the remaining five persons will not find a big enough shoe.

We claim that if k persons have left for some $1 \leq k \leq 4$, then one of the remaining $10 - k$ persons will find a big enough shoe. Let us denote the sizes of the shoes of the guests by $s_1, s_2, \ldots, s_{10}$. If the claim is false, then for some $1 \leq k \leq 4$, there are $10 - k$ guests whose shoes have sizes larger than some $10 - k$ numbers among $s_1, \ldots, s_{10}$. So, some $10 - k$ numbers among $s_1, \ldots, s_{10}$ are larger than some $10 - k$ numbers among $s_1, \ldots, s_{10}$. It follows that

$$10 - k + 10 - k \leq 10,$$

which shows that $k \geq 5$. This contradicts the bound $k \leq 4$. This proves the claim. Consequently, at most five guests may need to spend the night there.

We conclude that the largest number of the guests who might need to spend the night there is five. ■

Exercise 74 (RMO 2017a P4,). Consider n^2 unit squares in the xy-plane centred at the point (i, j) with integer coordinates for $1 \leq i \leq n, 1 \leq j \leq n$. It is required to colour each unit square in such a way that whenever $1 \leq i < j \leq n$ and $1 \leq k < \ell \leq n$, the three squares with centres at (i, k), (j, k), (j, ℓ) have distinct colours. What is the least possible number of colours needed?

Remark. For such problems, it is often useful to first work out a special case.

Walkthrough —

(a) First, work out a simple case in order to gain insight for the general case.

(b) One may consider the squares below a suitable diagonal.

(c) Extend to the general case!

First, let's work on it. Let us begin with the case $n = 8$. First, let us try to color the unit squares with as few colors as we can. This may provide some insight for the least number of colors required (for the case $n = 8$ and also possibly for the general case).

Note that any two unit squares at the bottom row of Fig. 20a have pairwise distinct colors. Let's apply the colors $1, \ldots, 8$ to these squares as in Fig. 20b. Note that in the second last row, all the unit squares except the first one, have colors different from those of the unit squares of the bottom row. Moreover, these seven unit squares have distinct colors. Let's apply the colors $3, \ldots, 9$ to these squares as in Fig. 20c. Similarly, in the third last row, the $8 - 3 + 1$ unit squares lying to the right, have colors different from those of the unit squares have been colored so far. Moreover, these $8 - 3 + 1$ unit squares have distinct colors. Let's apply another set of $8 - 3 + 1$ new colors (for example, $5, \ldots, 10$) to these squares as in Fig. 20d. One may continue this process to yield the coloring as in Fig. 20e.

In the remaining square in the second last row, we may use a color which has been used in that row, for instance, the color 2, as in Fig. 21a. In the third last row, the colors $3, 4$ may be used as in Fig. 21b. In the fourth last row, the colors smaller than 7 may be used as in Fig. 21c. One may continue this process to yield the coloring as in Fig. 21d. Note that the coloring as in Fig. 21d **does satisfy** the required condition. ♣

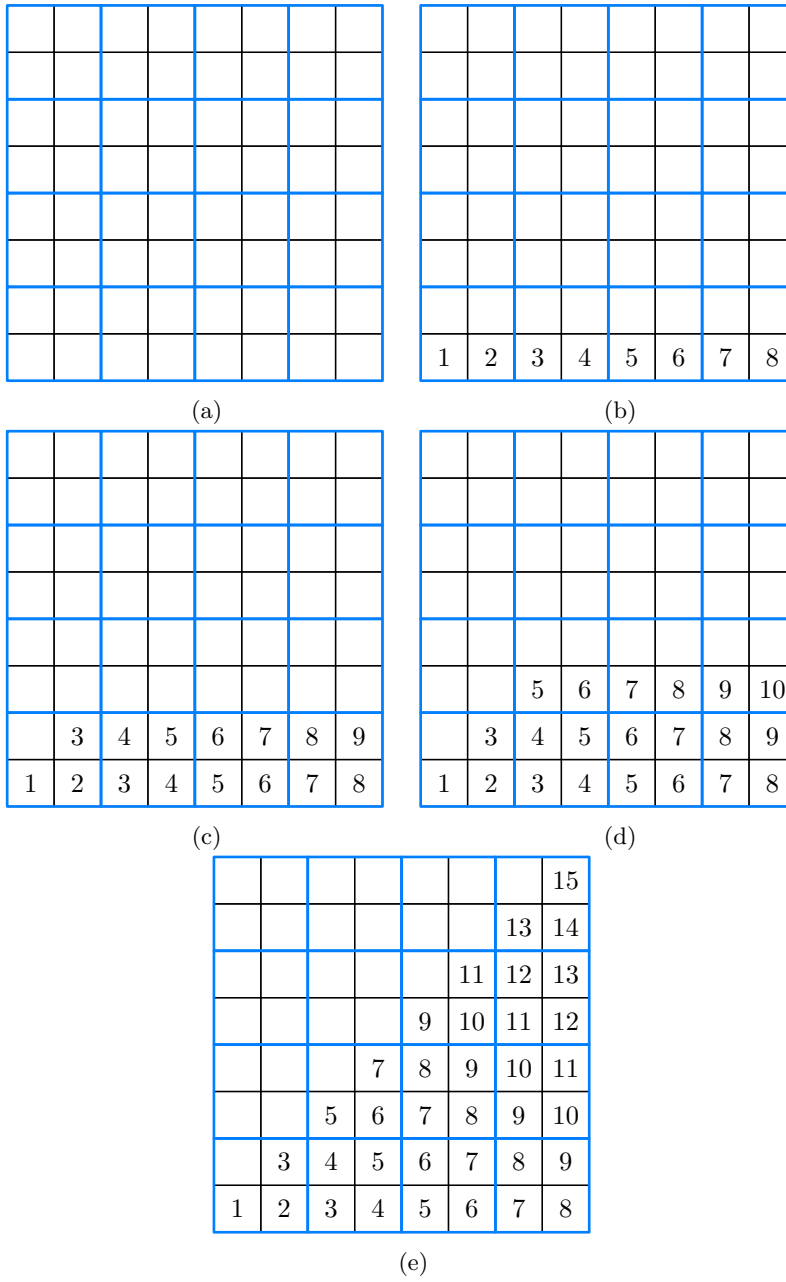
Remark. The **preceding argument alone** does not guarantee that the least possible number of colors have been used in Fig. 21d.

Skip this part for the first reading —

Remark. The **preceding argument alone** does not guarantee that the least possible number of colors have been used in Fig. 21d. It could have been the case that in our very first attempt of working out a special case, we might have proceeded in way which would have used more colors than what is required. For instance, consider the following approach.

In fact, that is what I (the author) did in a prior attempt, and then I could not prove that the number of colors that one may require to color the squares satisfying the required condition is at least as large as the number of colors used in the approach below. This led to think again, which yield the argument above.

¶ A prior attempt that used more colors than the minimum number of colors required. Let us begin with the case $n = 8$. First, let us try to color the unit squares with as few colors as we can. This may provide some insight for the least number of colors required (for the case $n = 8$ and also possibly for



(a) (b) (c) (d) (e)

Figure 20: India RMO 2017a P4, Exercise 74

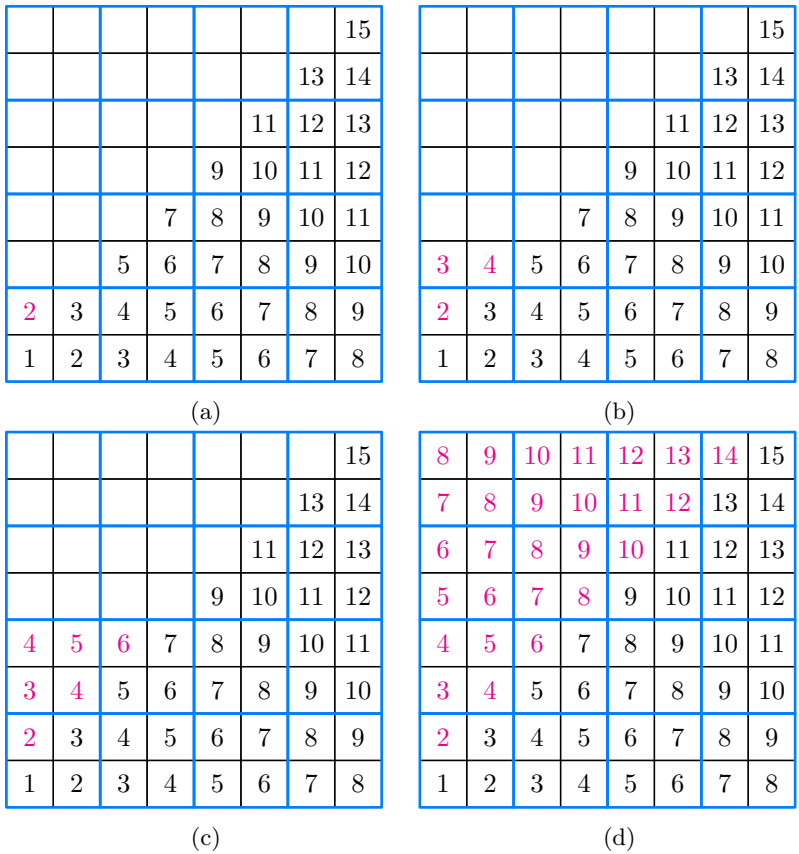


Figure 21: India RMO 2017a P4, Exercise 74

the general case).

Note that any two unit squares at the bottom row of Fig. 22a have pairwise distinct colors. Let us apply the colors $1, \ldots, 8$ to these squares as in Fig. 22b. Note that in the second last row, all the unit squares except the first one, have colors different from those of the unit squares of the bottom row. Moreover, these seven unit squares have distinct colors. Let us apply the colors $9, \ldots, 15$ to these squares as in Fig. 22c. Similarly, in the third last row, the $8-3+1$ unit squares lying to the right, have colors different from those of the unit squares have been colored so far. Moreover, these $8-3+1$ unit squares have distinct colors. Let us apply another set of $8-3+1$ new colors (for example, $16, \ldots, 21$) to these squares as in Fig. 22d. One may continue this process to yield the coloring as in Fig. 22e.

In the remaining square in the second last row, we may use a color which has been used in that row, for instance, the color 8, as in Fig. 23a. In the third last row, the colors $14, 15$ may be used as in Fig. 23b. In the fourth last row, the colors smaller than 22 may be used as in Fig. 23c. One may continue this process to yield the coloring as in Fig. 23d. Note that the coloring as in Fig. 23d does satisfy the required condition. ♣

Now that we have gained some experience, we may proceed to the general case as follows.

Bogus Solution. Suppose we have colored the n^2 unit squares using r colors so that the required condition is met. Consider the diagonal joining the upper right corner with the lower left corner. Note that the unit squares of the bottom row have distinct colors. Moreover, the unit squares lying to the right of the above-mentioned diagonal and contained in a row except the bottom one, have colors different from the unit squares contained in the rows below and lying to the right of that diagonal. Hence, we require at least

$$n + (n-1) + (n-2) + \cdots + 2 + 1 = \frac{1}{2}n(n+1)$$

colors.

Moreover, similar to the coloring as in Fig. 23, we may color the n^2 unit squares using $\frac{1}{2}n(n+1)$ colors so that required condition is satisfied.

This proves that the least number of colors required is $\frac{1}{2}n(n+1)$.

Question. Where does the above go wrong?

Now that we have gained some idea, we may proceed to the general case as follows.

Solution 69. If $n = 1$, then using one color works. Let us assume[1] that $n \geq 2$.

Suppose we have colored the n^2 unit squares using r colors so that the required condition is met. Consider the diagonal joining the upper right corner with the lower left corner. Note that the unit squares of the bottom row have

[1]The reason for considering the case $n = 1$ separately will become clear from the rest of the argument. Is it clear what would go wrong with the rest of the argument if $n = 1$?

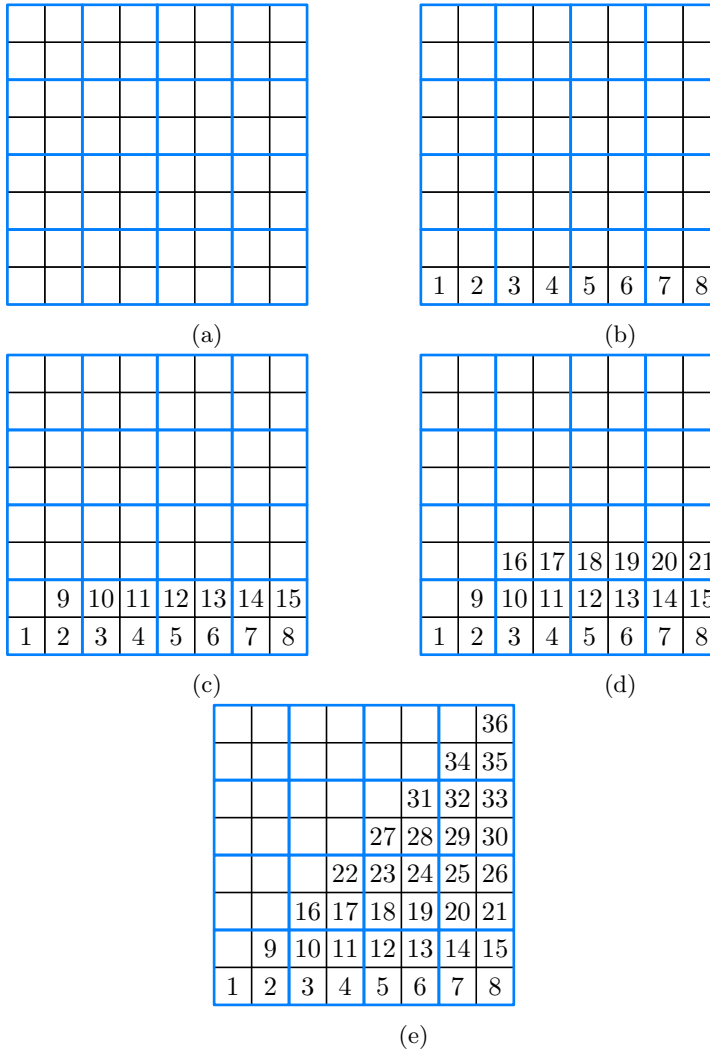


Figure 22: India RMO 2017a P4, Exercise 74

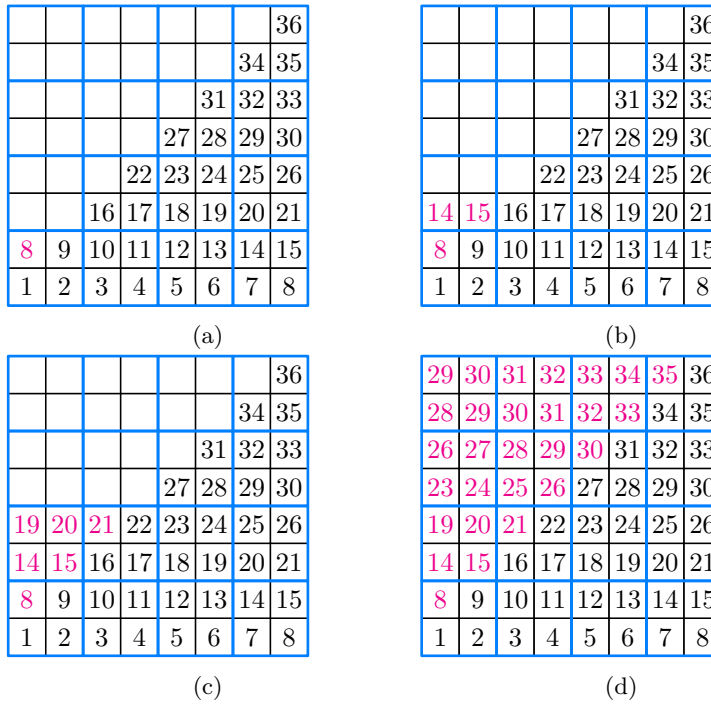


Figure 23: India RMO 2017a P4, Exercise 74

distinct colors. Moreover, the unit squares lying on the rightmost column except the bottom square, have colors different from those of the squares lying on the bottom row. Hence, we require at least

$$n + (n-1) = 2n - 1$$

colors.

Next, we show that there is a coloring of the n^2 unit squares using $2n-1$ colors satisfying the required condition. Indeed, if the colors $i, i+1, \ldots, i+n-1$ are applied to the squares lying in the i-th last row in an increasing order from the left to the right for all $1 \leq i \leq n$, then the squares in any row have distinct colors and so are the squares in any column, and hence, it is a coloring satisfying the required condition.

This shows that the least number of colors required is $2n-1$. ■

Exercise 75 (RMO 2016c P4,). There are 100 countries participating in an olympiad. Suppose n is a positive integer such that each of the 100 countries is willing to communicate in exactly n languages. If each set of 20 countries can communicate in at least one common language, and no language is common to all 100 countries, what is the minimum possible value of n?

Walkthrough —

(a) Choose a country C, and let L_1 denote a language in which C communicates. Note that L_1 is not common to some country C_1. Hence, some language L_2 is common to the countries C, C_1.

(b) The language L_2 is not common to some country C_2. Hence, some language L_3 is common to the countries C, C_1, C_2.

(c) **Continuing** this line of argument, show that there are pairwise distinct languages $L_1, \ldots, L_{20}$ and C communicates in them.

(d) Conclude that $n \geq 20$.

(e) Show that 20 is the minimum possible value of n.

Solution 70.

Claim — The inequality $n \geq 20$ holds.

Proof of the Claim. Let C denote a country. Let L_1 denote a language in which C communicates. By hypothesis, the language L_1 is not common to some country C_1. Some language L_2 is common to the countries C, C_1. By hypothesis, L_2 is not common to some country C_2. Note that some language L_3 is common to the countries C, C_1, C_2.

Suppose for some integer $2 \leq i$, there are countries $C_1, C_2, \ldots, C_i$ and there are languages $L_1, L_2, \ldots, L_{i+1}$ such that for any $1 \leq k \leq i$, the language L_{k+1}

is common to the $k+1$ countries $C, C_1, \ldots, C_k$, and L_k is not common to C_k. By the hypothesis, L_{i+1} is not common to some country C_{i+1}. If $i+2 \leq 20$, then the $i+2$ countries $C, C_1, \ldots, C_{i+1}$ have some common language L_{i+2}. By induction, there are countries $C_1, C_2, \ldots, C_{19}$, and there are languages $L_1, \ldots, L_{20}$ such that for any $1 \leq k \leq 19$, the language L_{k+1} is common to the $k+1$ countries $C, C_1, \ldots, C_k$, and that the language L_k is not common to the country C_k. For any $1 \leq k \leq 19$, the language L_{k+1} is common to the $k+1$ countries $C, C_1, \ldots, C_k$, and for any $1 \leq i \leq k$, the language L_i is not common to C_i, and hence, L_{k+1} is not equal to any of $L_1, \ldots, L_k$. This proves that the languages $L_1, \ldots, L_{20}$ are pairwise distinct, and C communicates in them. The Claim follows. □

Claim — One possible value of n is 20.

Proof of the Claim. Let $\mathscr{L}_1, \ldots, \mathscr{L}_{21}$ be pairwise languages. Let $\mathscr{C}_1, \ldots, \mathscr{C}_{100}$ be a choice of 100 countries. Suppose for $1 \leq i \leq 21$, the country $\mathscr{C}_i$ communicates in all the languages $\mathscr{L}_1, \ldots, \mathscr{L}_{21}$ other than $\mathscr{L}_i$. Assume further that for any $22 \leq i \leq 100$, the country $\mathscr{C}_i$ communicates in $\mathscr{L}_1, \ldots, \mathscr{L}_{20}$. Note that each country communicates in exactly 20 languages, and that no language is common to all the 100 countries. Moreover, each set of 20 countries among $\mathscr{C}_1, \ldots, \mathscr{C}_{21}$ share exactly one common language.

Note that at most 19 countries from $\mathscr{C}_1, \ldots, \mathscr{C}_{21}$, share at least two languages among $\mathscr{L}_1, \ldots, \mathscr{L}_{21}$ in common, and hence share at least one language among $\mathscr{L}_1, \ldots, \mathscr{L}_{20}$ in common. Hence, each set of 20 countries among $\mathscr{C}_1, \ldots, \mathscr{C}_{100}$, having at most 19 countries from $\mathscr{C}_1, \ldots, \mathscr{C}_{20}$, share at least one language in common. This proves the Claim. □

Combining the above Claims, it follows that the minimum possible value of n is 20. ■

Exercise 76 (RMO 2018b P4,). Suppose 100 points in the plane are coloured using two colours, red and white, such that each red point is the centre of a circle passing through at least three white points. What is the least possible number of white points?

Summary — It relies on the fact that one can find enough points on the plane such that no three of them are collinear and no four of them are concyclic.

Walkthrough —

(a) There is an upper bound on the number of the red points in terms of the number of the white points. This gives an upper bound on the total number of points, which is 100, in terms of the number of the white points.

(b) Use this bound to guess the least possible number of the white points, which would turn out to be 10.

(c) Begin with 10 white points on the plane in *general position*, and then, introduce enough red points to construct a configuration of 100 points with the stated properties.

Solution 71. Let n denote the number of white points. Since each red point is the centre of a circle passing through at least three white points, it follows that the number of red points is at most $\binom{n}{3}$. This shows that

$$n + \binom{n}{3} \geq 100.$$

Note that $n \mapsto n + \binom{n}{3}$ defines an increasing function on the nonnegative integers. Observe that

$$9 + \binom{9}{3} = 93, \quad 10 + \binom{10}{3} = 130.$$

This implies that $n \geq 10$.

We claim that there is a configuration of 100 points on the plane such that it admits a coloring using two colors, red and white, such that precisely 10 points are colored white, and that each red point is the centre of a circle passing through at least three white points. Indeed, consider 10 points on the plane such that no three of them are collinear and no four of them are concyclic [2]. Color these 10 points white. These white points have $\binom{10}{3} = 120$ subsets of size 3. Consider only 90 such subsets of the white points, and for any such subset of size 3, color the center of the circle passing through them red. Since no three white points are collinear and no four white points are concyclic, it follows that there are precisely 90 pairwise distinct red points. So, the red and the white points together form a set of 100 points such that each red point is the centre of a circle passing through at least three white points. ■

Exercise 77 (INMO 2025 P5, proposed by Pranjal Srivastava and Rohan Goyal)**.** Greedy goblin Griphook has a regular 2000-gon, whose every vertex has a single coin. In a move, he chooses a vertex, removes one coin each from the two adjacent vertices, and adds one coin to the chosen vertex, keeping the remaining coin for himself. He can only make such a move if both adjacent vertices have at least one coin. Griphook stops only when he cannot make any more moves. What is the maximum and minimum number of coins that he could have collected?

[2] Why does such a collection exist? This could be intuitively clear, but can you write down a precise proof? Does induction help?

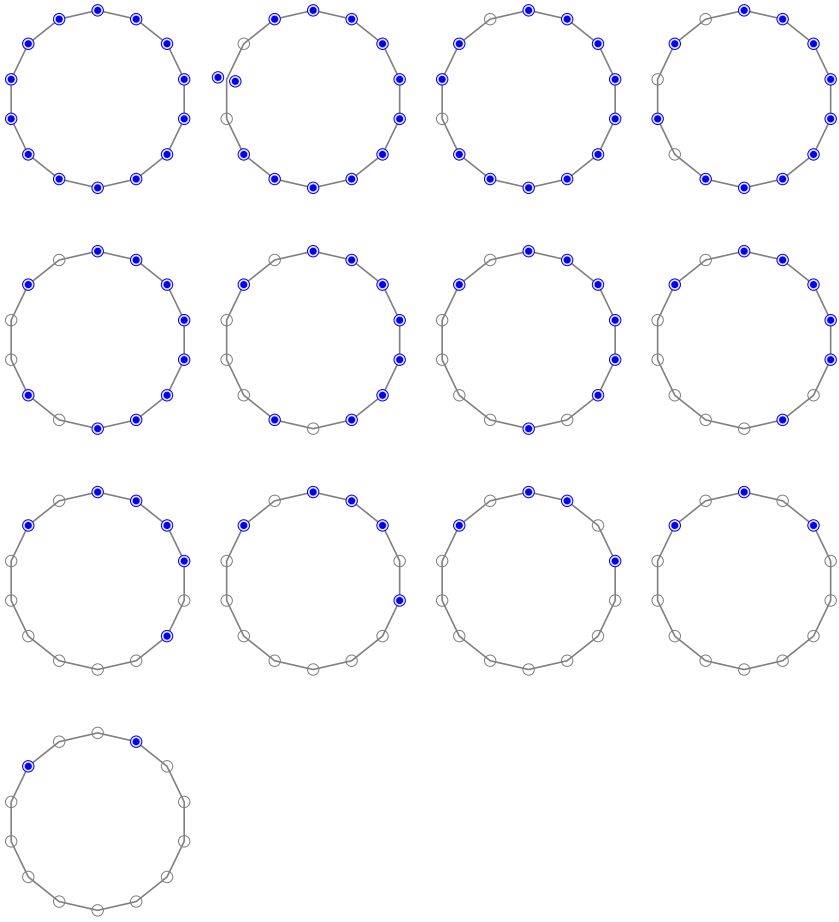


Figure 24: INMO 2025 P5, Collecting 12 coins, Exercise 77

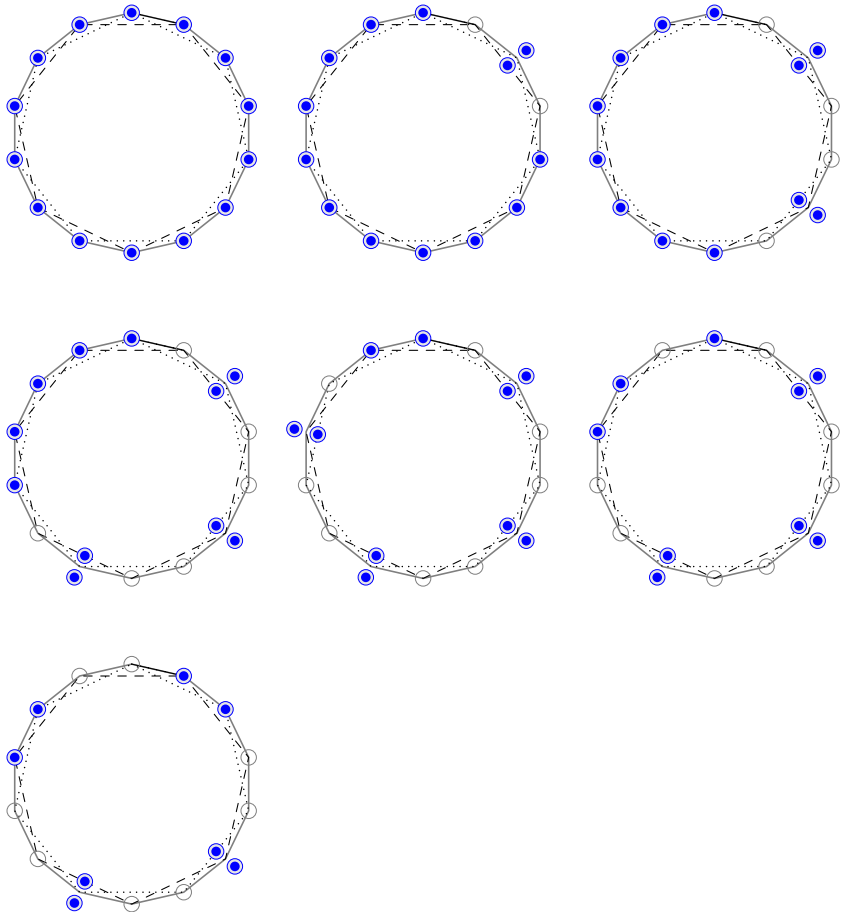


Figure 25: INMO 2025 P5, Collecting 6 coins, Exercise 77

Walkthrough —

(a) Show that 1998 coins can be collected (see Fig. 24).

(b) Show that the difference of the number of coins at the vertices with odd indices, and the number of coins at the vertices with even indices, does not change modulo 3 after any move.

(c) Conclude that no more than 1998 coins can be collected.

(d) Show that 668 coins can be collected.

(e) Show that at least 668 coins are collected when Griphook stops.

The following is from AoPS, and is due to Rohan Goyal.

Solution 72. Denote the regular 2000-gon by $A_1A_2\dots A_{2000}$. The claims below imply that the maximum (resp. minimum) number of coins that Griphook could have collected is 1998 (resp. 668).

Claim — No more than 1998 coins can be collected.

Proof of the Claim. Note that after any move, the number of coins at the vertices with odd indices and the number of coins at the vertices with even indices either reduces by 2 and increases by 1 respectively, or increases by 1 and reduces by 2 respectively. This shows that the difference of the number of coins at the vertices with odd indices and the number of coins at the vertices with even indices, does not change modulo 3 after any move, and hence, this difference is zero modulo 3 after any move.

Since the number of coins remaining after any move is at least 1, there are at least two coins left at the end of any move. Consequently, no more that 1998 coins can be collected. □

Claim — There is a way to collect 1998 coins.

Proof of the Claim. Let us denote the number of coins at the vertices of the regular 2000-gon by a string of length 2000, consisting of nonnegative integers, where the i-th entry of the string denotes the number of coins at the i-th vertex A_i.

In the following, 1^m (resp. 0^m) denotes a string of m 1's (resp. 0's). Note that by making a move, the string 1^{2000} can be turned to $1^{1996}0201$, which can be turned to $1^{1995}01010$ by making another move. Observe that the sub-string 101 can be turned to 010 by making a move. Hence, making enough suitable moves, the string $1^{1995}01010$ can be reduced to $010^{1996}10$. While reaching $010^{1996}10$ from 1^{2000}, a total of 1998 coins have been earned, proving the claim. □

Claim — There is a sequence of moves, which leads to the collection of 668 coins when Griphook stops.

Proof of the Claim. Write the string 1^{2000} as $1(111)^{666}1$, which can be turned to $1(020)^{666}1$, which is equal to $1020(020)^{664}0201$. This can be reduced to $0110(020)^{664}0110$. Note that no further move is possible, and hence Griphook has to stop, and he earned $2000 - 4 - 2 \times 664 = 668$ coins. □

Claim — For any sequence of moves, Griphook will have at least 668 coins after he stops.

Proof of the Claim. Let c_i denote the number of coins at the i-th vertex. Note that each move reduces the sum

$$S = \sum_{i=1}^{2000} \min\{c_{i-1}, c_{i+1}\}.$$

In this sum, the indices are to be read modulo 2000. In each move, this sum reduces at least by 1, and Griphook stops precisely when this sum reduces to 0. Note that at any moment, for any two one-vertex apart vertices, one of them contains at most one coin[3], that is,

$$\min\{c_{i-1}, c_{i+1}\} \leq 1$$

for any $1 \leq i \leq 2000$. Write

$$E = \sum_{1 \leq i \leq 2000, 2 \mid i} \min\{c_{i-1}, c_{i+1}\},$$

$$O = \sum_{1 \leq i \leq 2000, 2 \nmid i} \min\{c_{i-1}, c_{i+1}\}.$$

Note that in each move, one of E, O is reduced at most by 3. Hence, at least 334 moves are required to reduce any of E, O to 0. Since in each move, only one of E, O gets reduced, it follows that Griphook makes at least $334 + 334 = 668$ moves before he stops.

□

■

Exercise 78 (Moscow Mathematical Olympiad 2025 Grade 9 P4, proposed by A. Zaslavsky)**.** Each cell of a 100×100 square is painted either white or black. It turns out that each white cell has exactly two adjacent cells painted white, and each black cell has exactly two adjacent cells painted black. Find the maximum possible number of black cells.

[3] Why?

Walkthrough —

(a)

Solution 73.

■

Exercise 79 (All-Russian Mathematical Olympiad 2025 Grade 9 Day 2 P6, , proposed by S. L. Berlov)**.** Petya chooses 100 pairwise distinct positive numbers less than 1 and arranges them in a circle. In one operation, he may take three consecutive numbers a, b, c (in this order) and replace b with $a-b+c$. What is the greatest value of k such that Petya could initially choose the numbers and perform several operations so that k of the resulting numbers are integers?

Walkthrough —

(a)

Solution 74. Let $a_1, a_2, \ldots, a_{100}$ be the numbers on the circle in the clockwise order, prior to an operation. Consider the list of numbers, formed by taking the differences of consecutive numbers in the circle:

$$a_1 - a_2, a_2 - a_3, \ldots, a_{100} - a_1.$$

We call this list the *difference list* of the numbers on the circle. Petya chooses a_{i-1}, a_i, a_{i+1} (indices are taken modulo 100) and replaces a_i with $a_{i-1}-a_i+a_{i+1}$. Then the difference list changes as follows:

$$a_{i-1} - a_i, a_i - a_{i+1}$$

is replaced with

$$a_{i-1} - (a_{i-1} - a_i + a_{i+1}) = a_i - a_{i+1}, (a_{i-1} - a_i + a_{i+1}) - a_{i+1} = a_{i-1} - a_i.$$

In other words, the difference list remains unchanged, except that the two numbers $a_{i-1} - a_i$ and $a_i - a_{i+1}$ are swapped. Therefore, the difference list remains invariant up to reordering under the operations.

Since Petya chooses 100 pairwise distinct positive numbers less than 1, the difference list consists of 100 nonzero numbers less than 1 in absolute value. So, after performing several operations, no two adjacent numbers on the circle can be integers, and hence, at most 50 of the numbers on the circle can be integers. This shows that $k \leq 50$.

To show that $k = 50$ is achievable, Petya can initially place the numbers $c_1, c_2, \ldots, c_{100}$ on the circle in the clockwise order, where

$$c_i = \begin{cases} 1 - \frac{1}{2^i} & \text{if } i \text{ is even,} \\ 1 - \frac{1}{2^{i-1}} - \frac{1}{2^{i+1}} & \text{if } i \text{ is odd and } i \neq 1, \\ 1 - \frac{1}{2^{100}} - \frac{1}{2^2} & \text{if } i = 1. \end{cases}$$

Note that operating on c_i for all odd i, leads to having 1 at all odd positions, and hence 50 integers on the circle. This proves that $k = 50$ is achievable.

Hence, the greatest value of k is 50. ■